ADVANCED ROBOTIC PERCEPTION / IMAGE PROCESSING AND COMPUTER VISION

ROBUST FEATURES, SELECTION AND REDUCTION

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Agenda

- Feature point extractors and descriptors
- Feature selection
- Dimensionality reduction

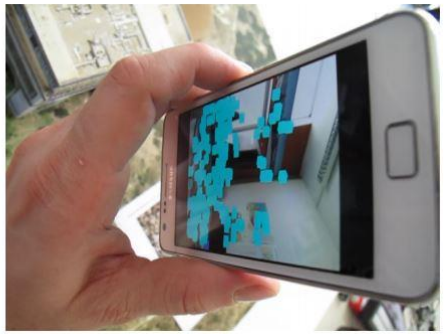


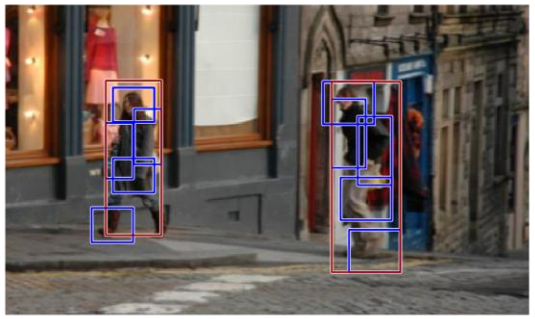
Features

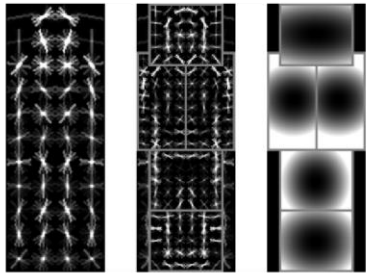
- Feature descriptors: Describing local regions
  - Also just called features

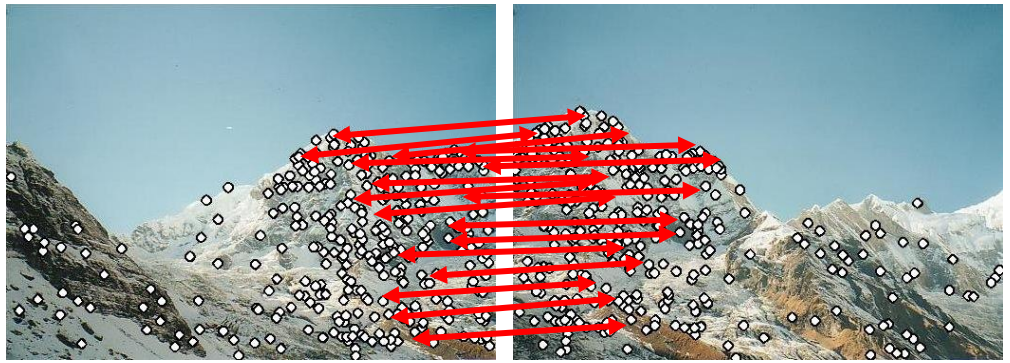
- Useful for:
  - Matching
  - Recognition
  - Detection
  - …

















Features

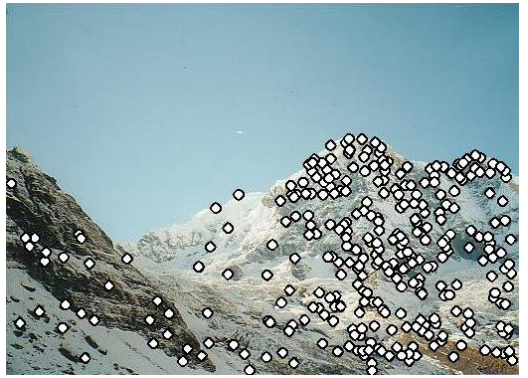
Objectives:

- Want to find patches in image that are useful or have some meaning
- For object detection, want a patch that is common to that object but not in general
- For panorama stitching, want patches that we can find easily in another image of same place
- Good features are unique!
  - Can find the “same” feature easily
  - Not mistaken for “different” features



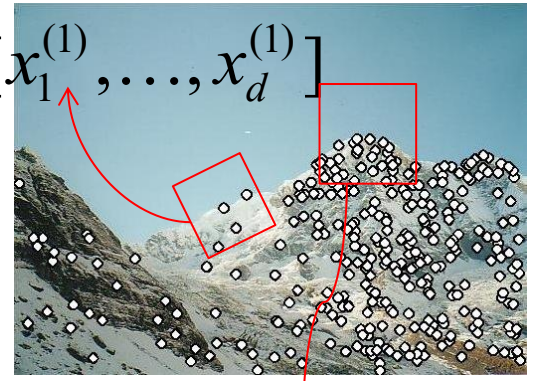
Features – main components

1) **Detection**: Identify the interest points



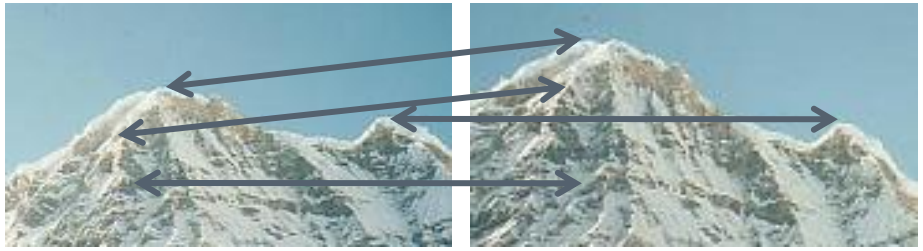
2) **Description**: Extract vector feature descriptor surrounding each interest point.

$$\mathbf{x}_1 = [x_1^{(1)}, \ldots, x_d^{(1)}]$$



$$\mathbf{x}_2 = [x_1^{(2)}, \ldots, x_d^{(2)}]$$

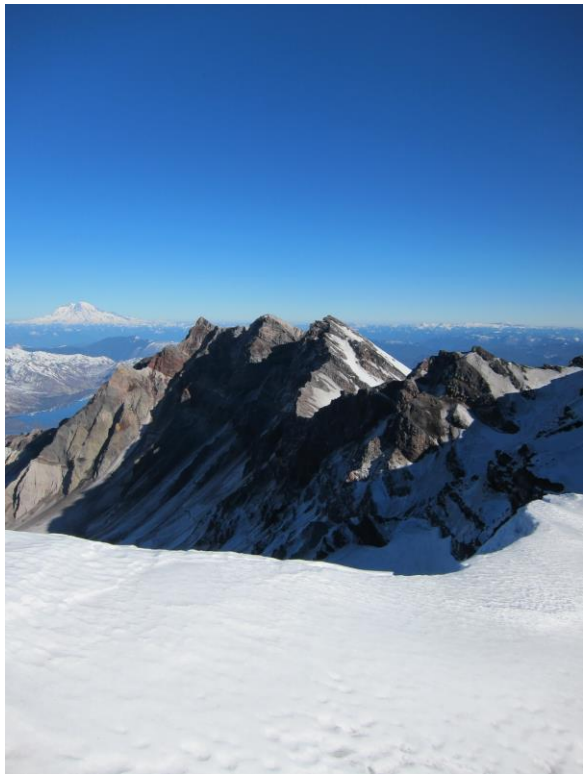
3) **Matching**: Determine correspondence between descriptors in two views or between model and image

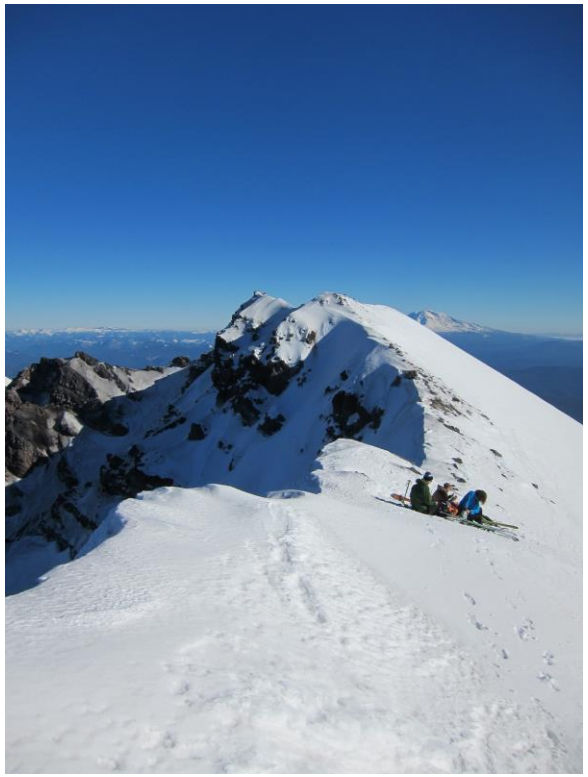


Features

Application example:

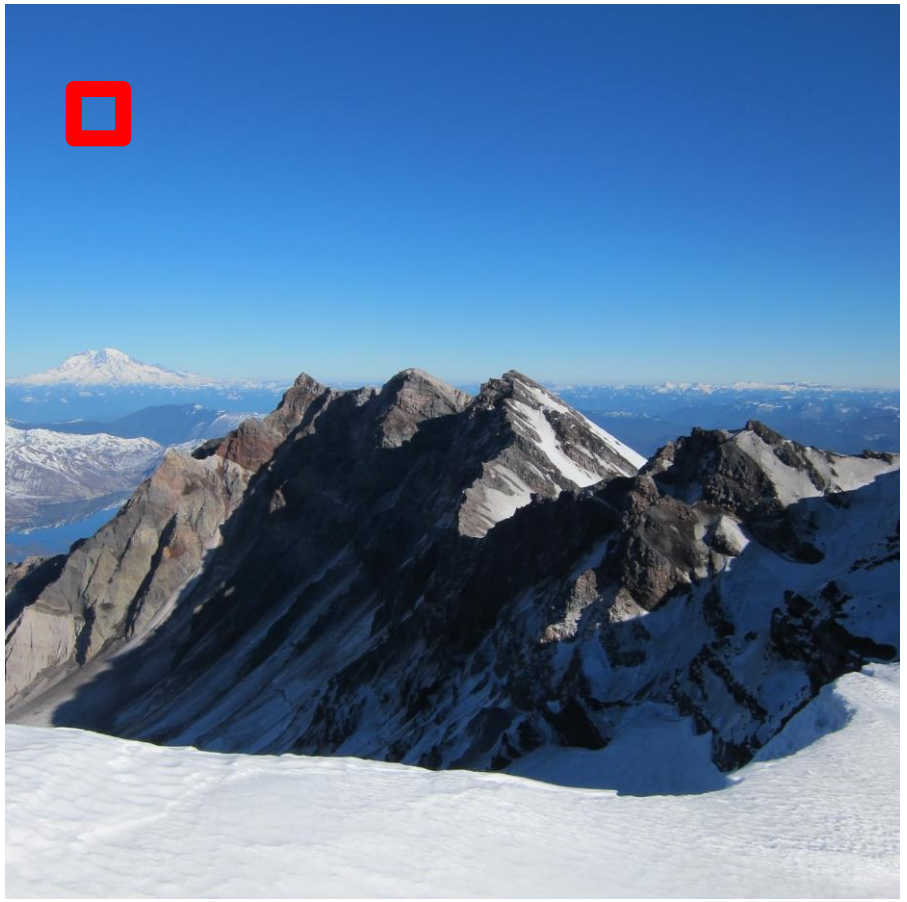
- Stitching a panorama: Looking for patches in image to match other image





Features

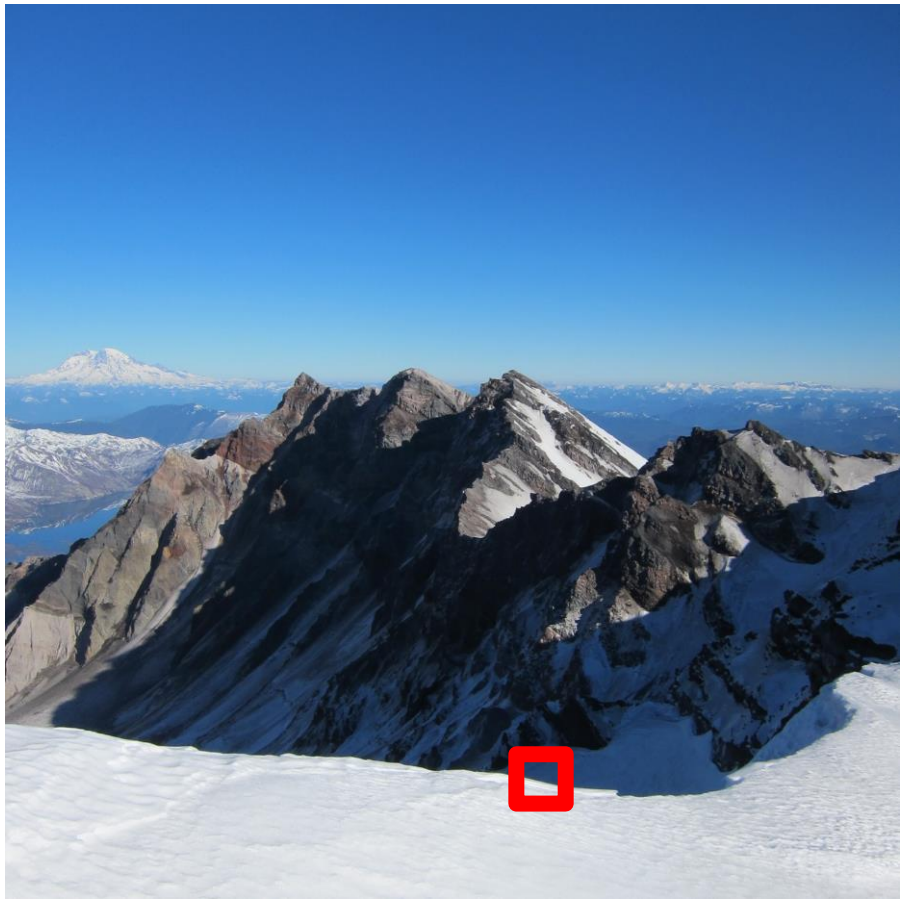
- Unique patches?
- Sky: bad
  - Very little variation
  - Could match any other sky



Features

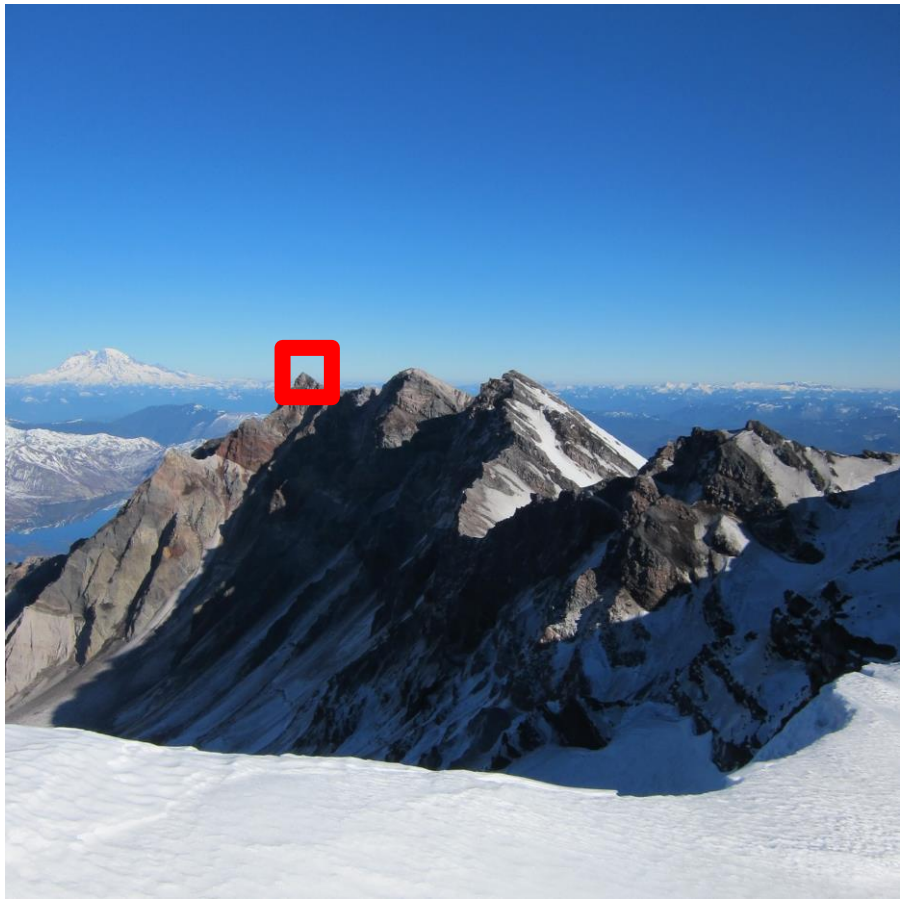
- Unique patches?
- Sky: bad
  - Very little variation
  - Could match any other sky

- Edge: ok
  - Variation in one direction
  - Could match other patches along same edge



Features

- Unique patches?
- Sky: bad
  - Very little variation
  - Could match any other sky
- Edge: ok
  - Variation in one direction
  - Could match other patches along same edge
- Corners: good!
  - Only one alignment matches



Features

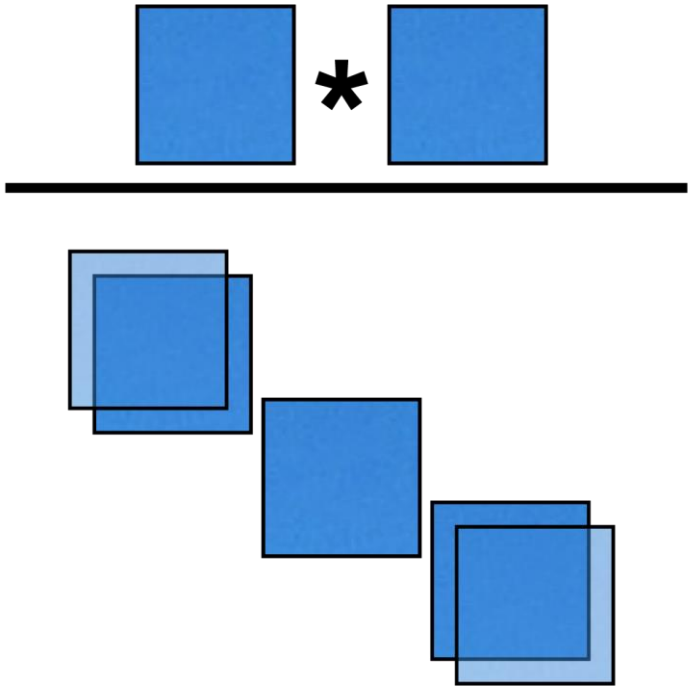
- How to find the unique patches?

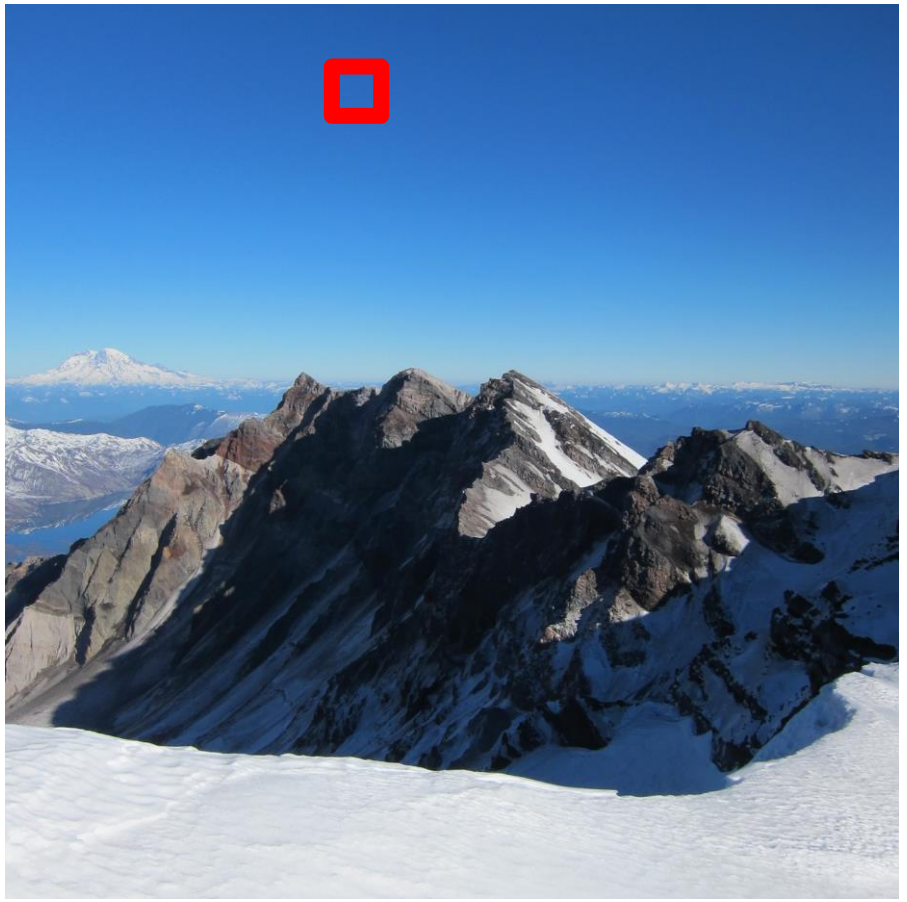
  a) Distance (difference) between patch and every other patch
     - Computationally heavy!

  b) Auto-correlation: How well does the image patch match a shifted version of itself?
     - Measure of self-difference
     - $\sum_d \sum_{x,y} (I(x,y) - I(x+dx, y+dy))^2$

Features

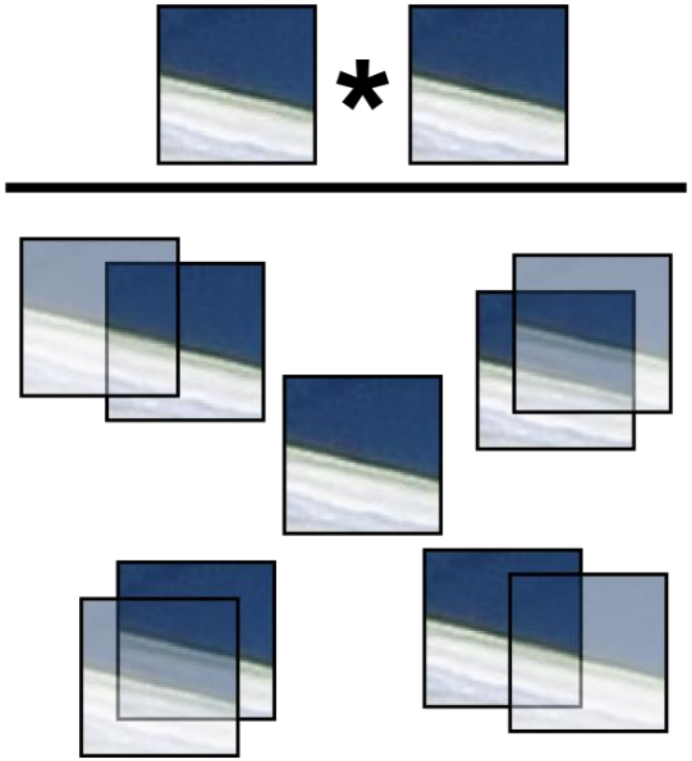
- Sky: Self-difference low everywhere

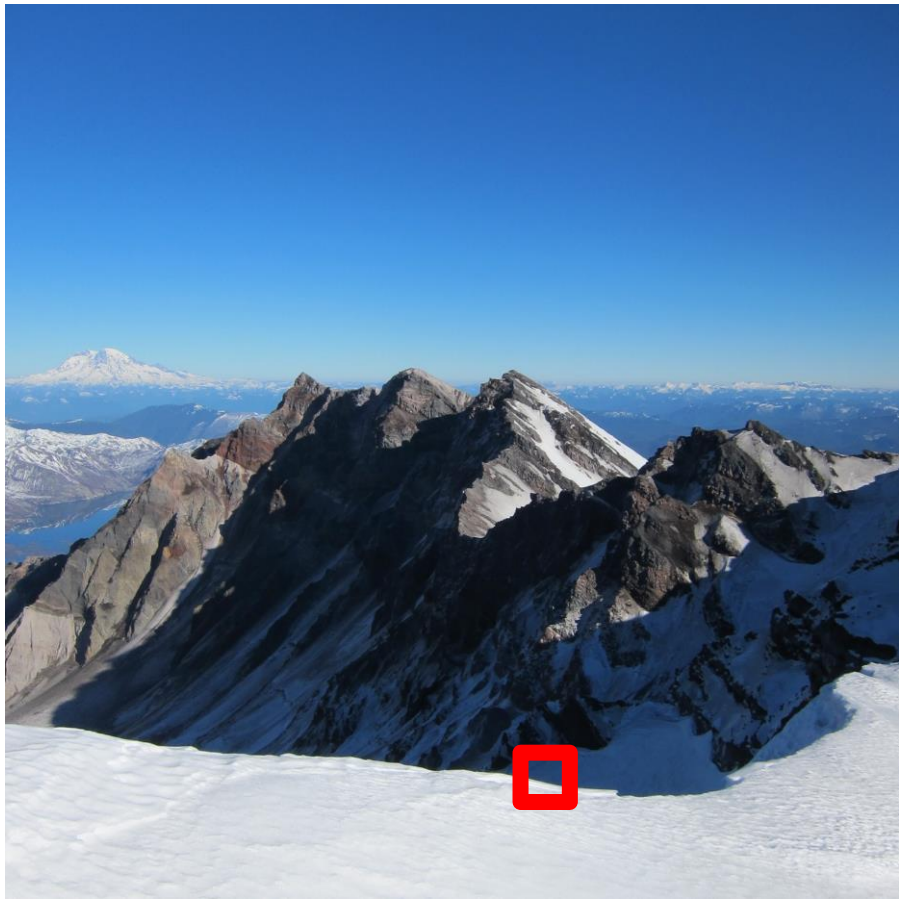




Features

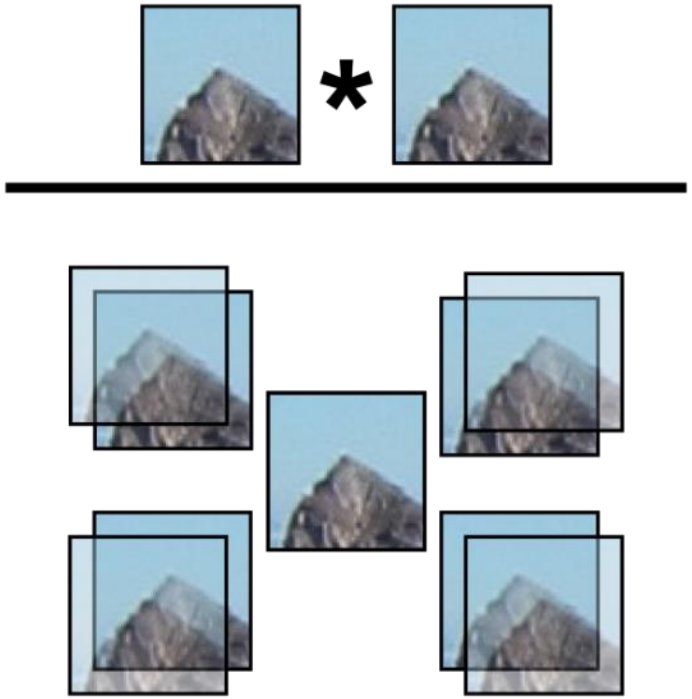
- Edge: Self-difference low along edge

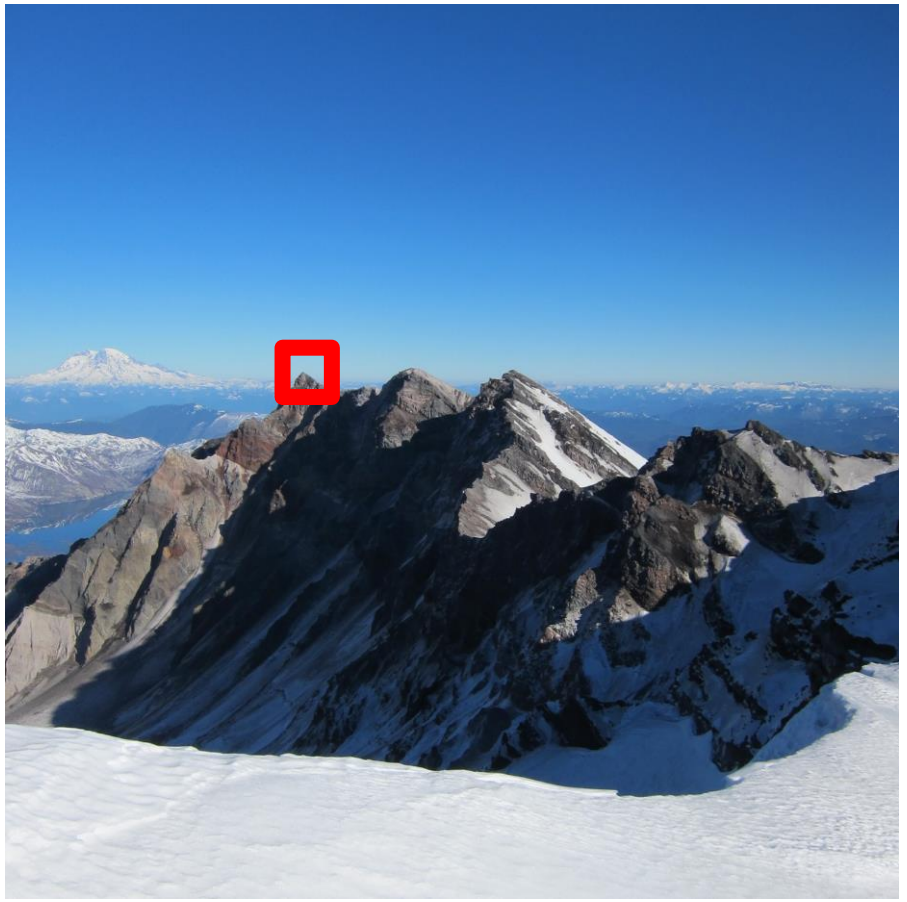




Features

- Corner: Self-difference mostly high





Features

- How to find the unique patches?

  a) Distance between patch and every other patch
     - Computationally heavy!

  b) Auto-correlation: How well does the image match a shifted version of itself?
     - Measure of self-difference
     - $\sum_d \sum_{x,y} (I(x,y) - I(x+dx, y+dy))^2$
     - Still computationally expensive – need an approximation

  c) Gradients, x- and y-direction
     - Zero: plain image patch
     - One direction: Edge
     - Two directions: Corner, good patch!

Features

- Analysis of gradients: Eigen vectors/- values!
- Need structure matrix for patch, just a weighted sum of nearby gradient information

- $$S_w[p] = \begin{bmatrix} \sum_r w[r]\,(I_x[p-r])^2 & \sum_r w[r] I_x[p-r] I_y[p-r] \\ \sum_r w[r] I_x[p-r] I_y[p-r] & \sum_r w[r]\big(I_y[p-r]\big)^2 \end{bmatrix}$$

- Not as complex as it looks, weighted sum of gradients near pixel
  - Can use Gaussian weighting

Features

$$S_w[p] = \begin{bmatrix} \sum_r w[r]\,(I_x[p-r])^2 & \sum_r w[r] I_x[p-r] I_y[p-r] \\ \sum_r w[r] I_x[p-r] I_y[p-r] & \sum_r w[r]\big(I_y[p-r]\big)^2 \end{bmatrix}$$

- Eigen vectors/values of this matrix summarize the distribution of the gradients nearby
  - λ_1 and λ_2 are eigenvalues
  - λ_1 and λ_2 both small: no gradient
  - λ_1 >> λ_2: gradient in one direction
  - λ_1 and λ_2 similar: multiple gradient directions, corner

Harris Corner Detection

- Calculate derivatives I_x and I_y
- Calculate $I_x I_x$, $I_y I_y$, $I_x I_y$
- Calculate weighted sums (S)
- Estimate response based on smallest eigenvalue
- Non-maximum suppression (like Canny)

- Approximation of smallest eigenvalue:

$$\lambda_{min} \approx \frac{\lambda_1 \lambda_2}{(\lambda_1 + \lambda_2)} = \frac{\det(S)}{trace(S)}$$

or

$$R = \det(S) - k\big(trace(S)\big)^2 = \lambda_1 \lambda_2 - k(\lambda_1 + \lambda_2)^2$$

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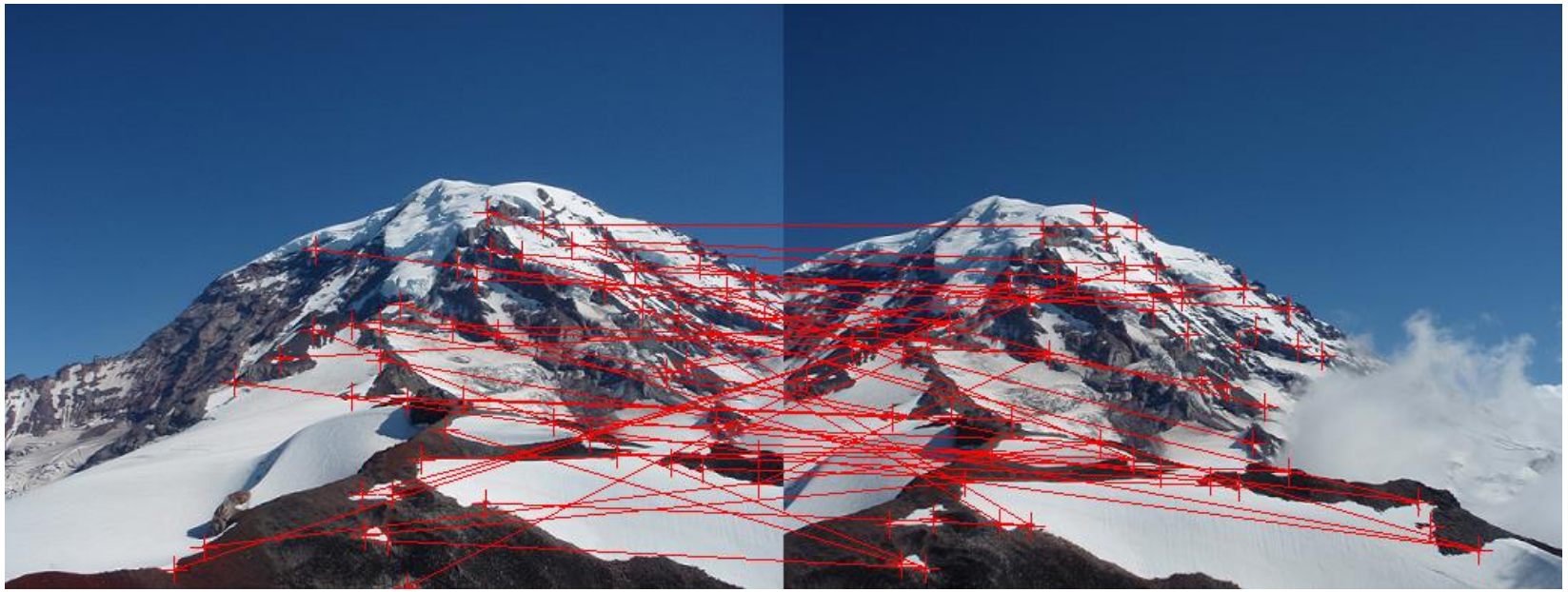
C. Harris and M. Stephens: A combined corner and edge detector. Proceedings of the 4th Alvey Vision Conference, pages 147-151, 1988





Harris Corner Detection

- Matching patches: Use extracted descriptors
- Loop over good patches in one image and find best match in other image
- Do something – transformation, detection, tracking, etc…



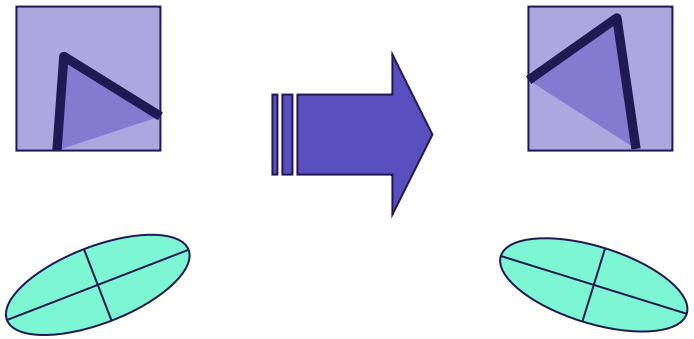
Robust features

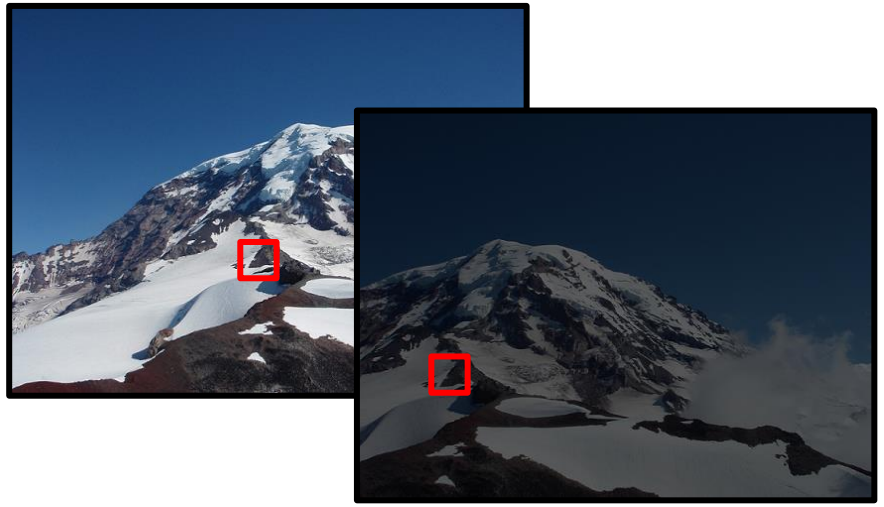
Harris Corner detection: Good, but issues remain!

Properties:

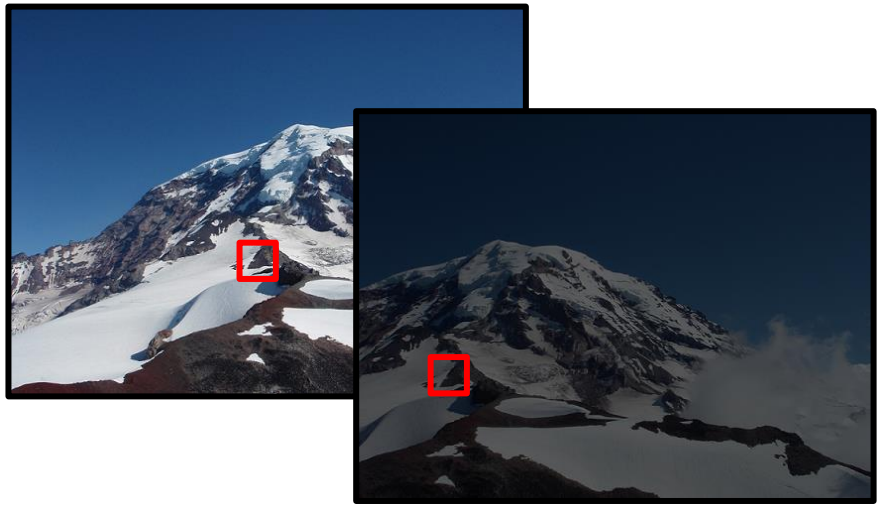
- Rotation invariant

- Ellipse rotates but its shape (i.e. eigenvalues) remains the same





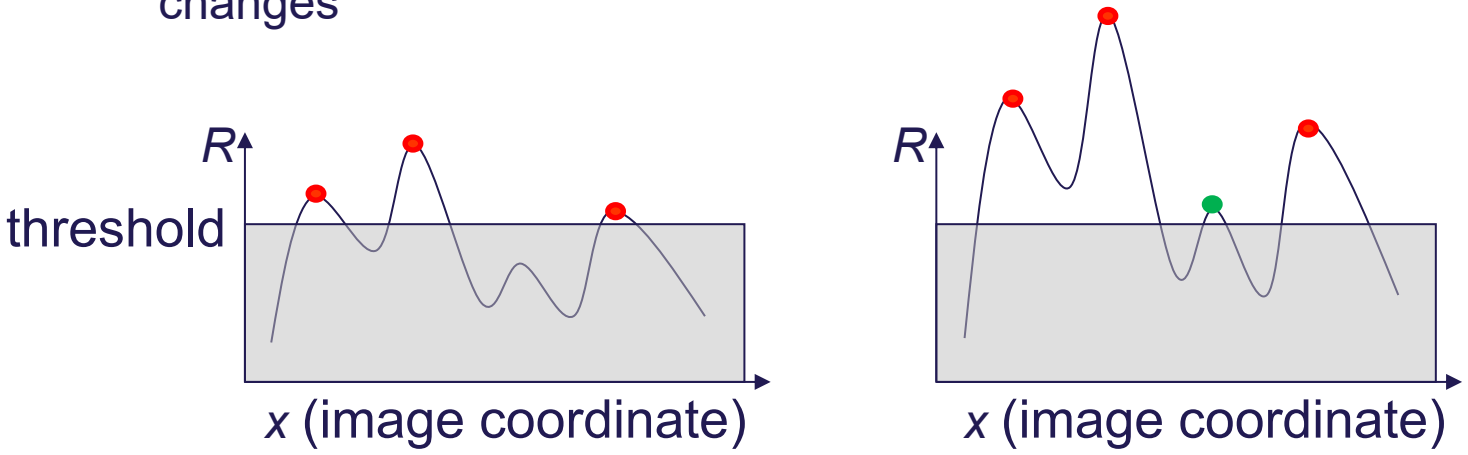
Robust features



Harris Corner detection: Good, but issues remain!

Properties:

- Invariance to image intensity change?
  - Only derivatives are used: invariance to intensity shift $I \rightarrow I+b$
  - Intensity scale: $I \rightarrow aI$. Because of fixed intensity threshold on local maxima, only partial invariance to multiplicative intensity changes

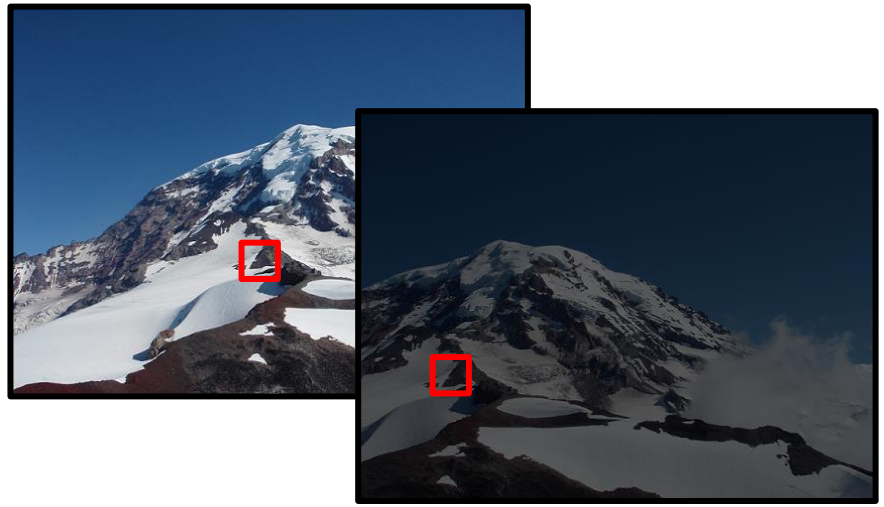


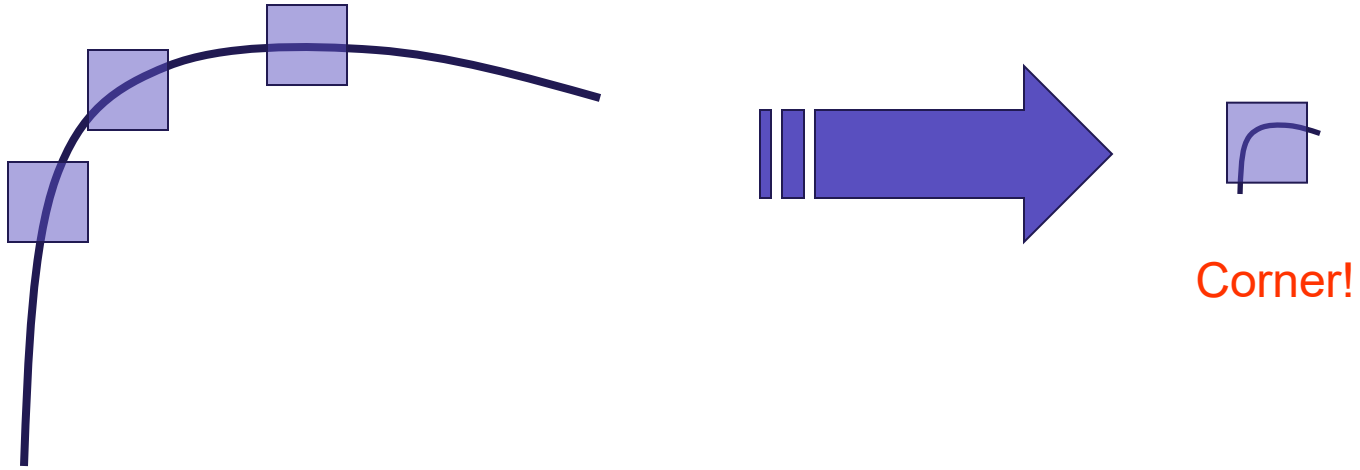
Robust features

Harris Corner detection: Good, but issues remain!

Properties

- Not invariant to image scale





Corner!

All points will be classified as edges

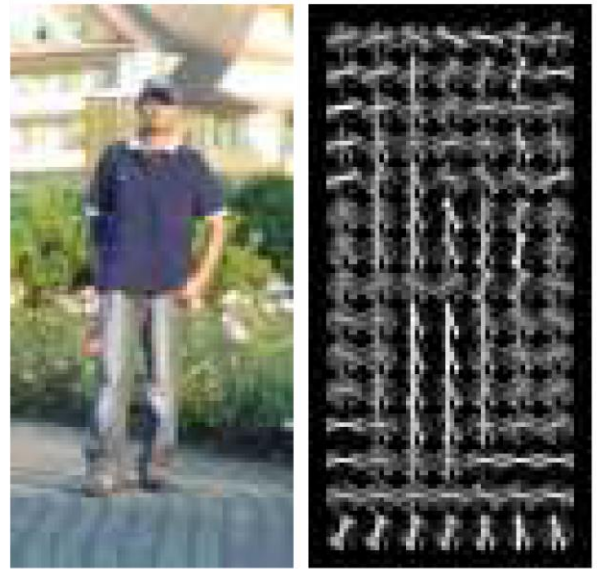
Histogram of Oriented Gradients (HOG)

Idea: Objects can be described by the distribution of local intensity gradients (edges)

Originally proposed for human detection

Algorithm:

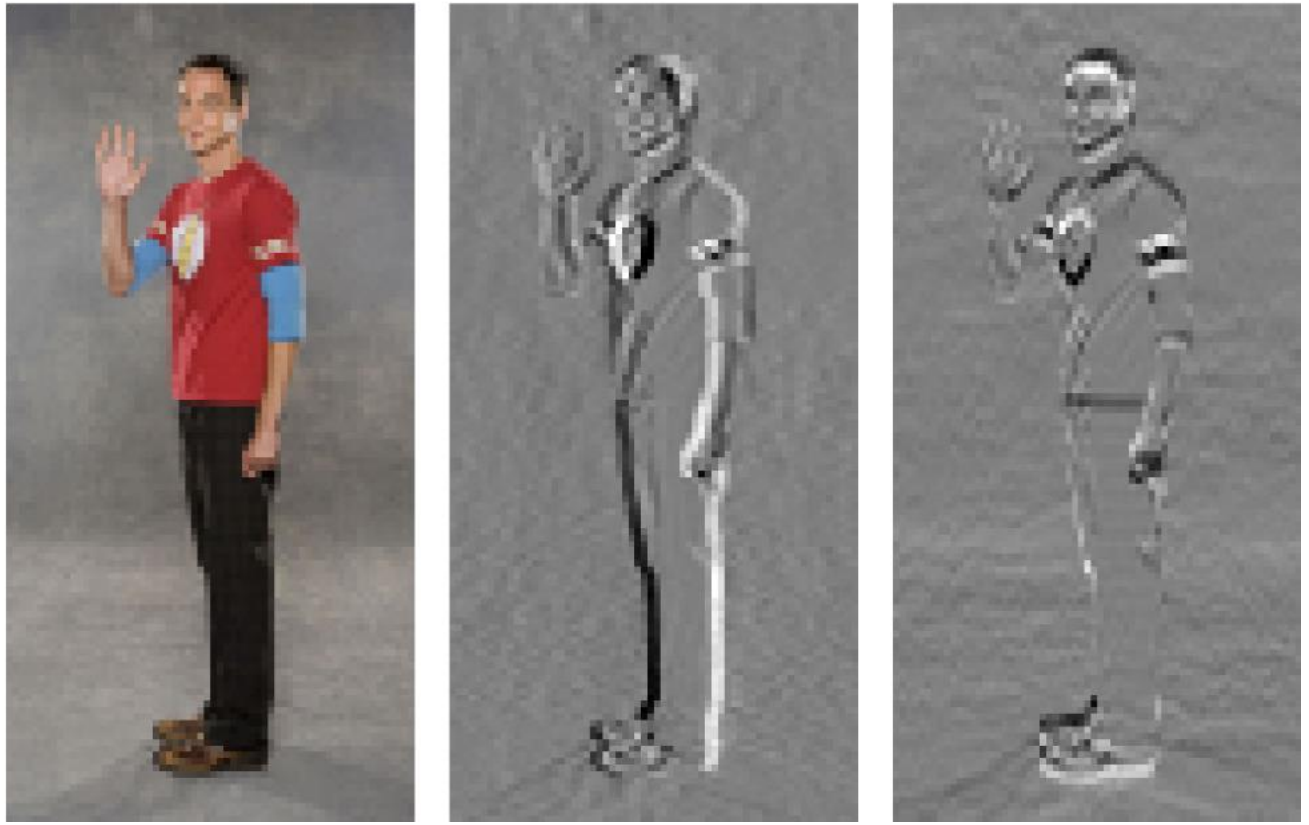
1. Normalize colour
2. Compute gradients
3. Weighted vote into spatial and orientation cells
4. Contrast normalize over overlapping spatial blocks
5. Collect HOG's over detection window
6. *Run linear SVM classifier*



Dalal and Triggs: "Histogram of Oriented Gradients for Human Detection". CVPR 2005

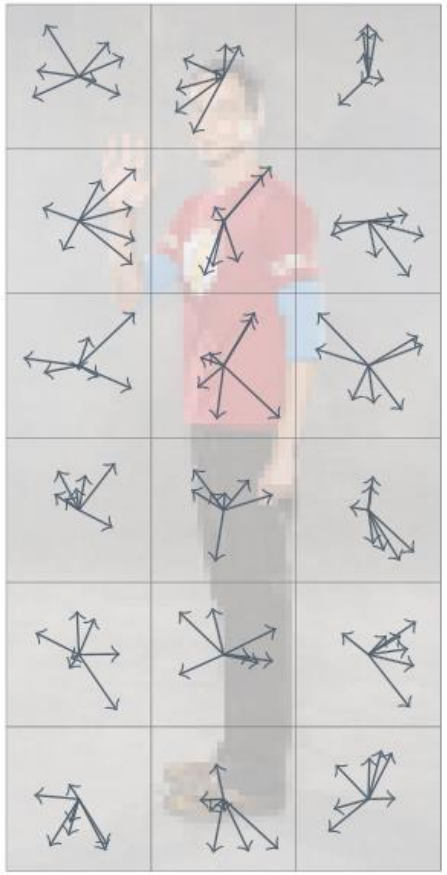
Histogram of Oriented Gradients (HOG)

- Compute gradients
- Kernels: $[-1, 0, 1]$ $[-1, 0, 1]^T$



Histogram of Oriented Gradients (HOG)

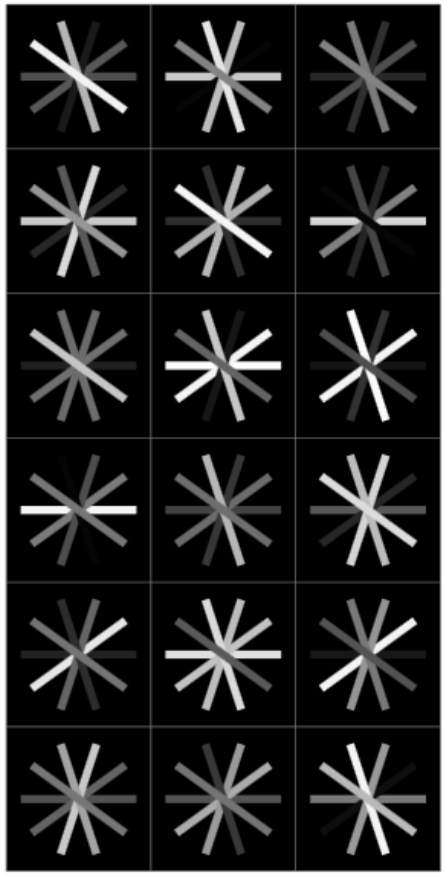
Compute gradient orientations in cells



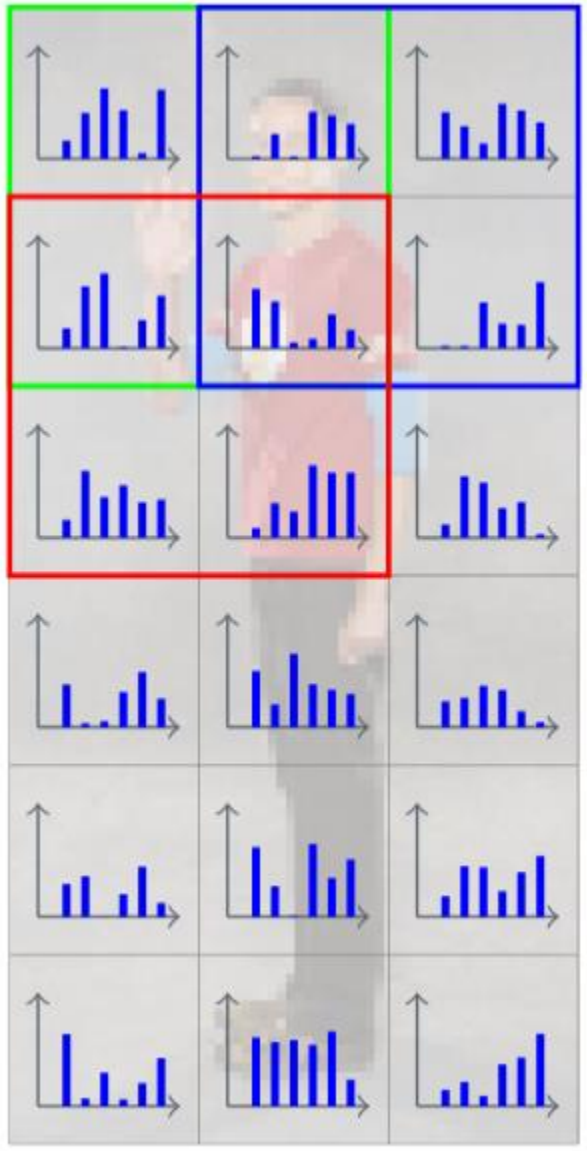
Assemble histograms



Visualized



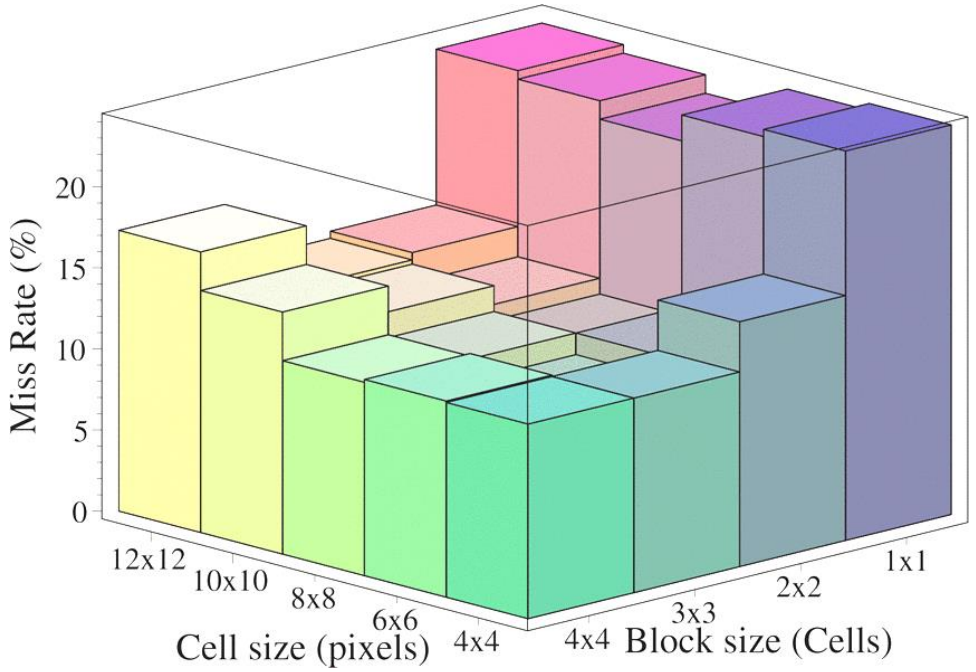
Histogram of Oriented Gradients (HOG)



- Normalization is done in (overlapping) blocks, which span multiple cells
- L2-norm, including a regularization term ϵ:

$$\boldsymbol{v} \rightarrow \boldsymbol{v}/(\|\boldsymbol{v}\|_2^2 + \epsilon^2)$$

where **v** is the unnormalized combined block feature vector



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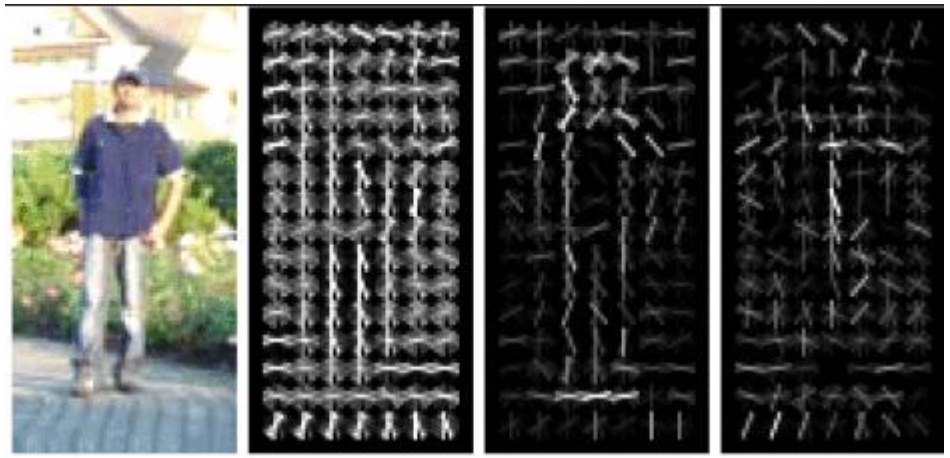
Histogram of Oriented Gradients (HOG)

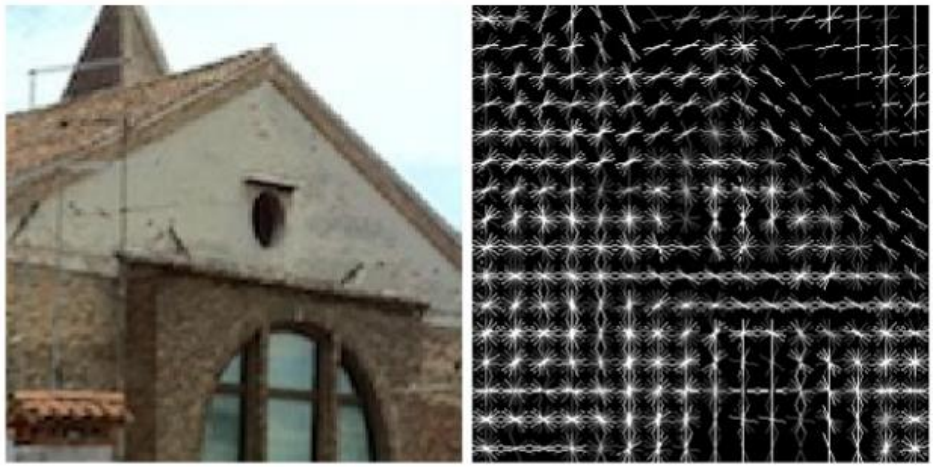
Idea: Objects can be described by the distribution of local intensity gradients (edges)

Originally proposed for human detection

Algorithm:

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2. Compute gradients
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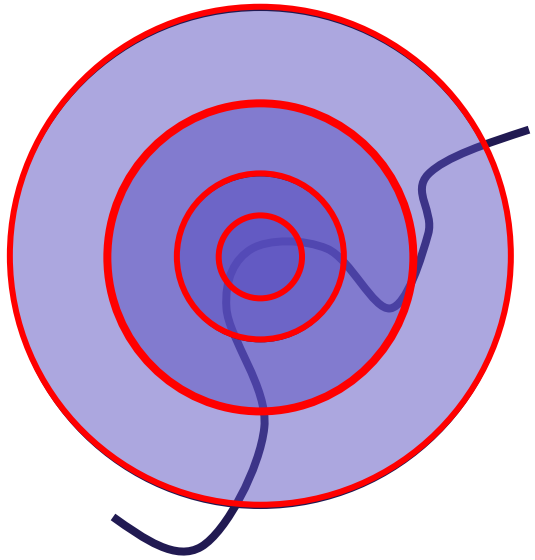
Dalal and Triggs: "Histogram of Oriented Gradients for Human Detection". CVPR 2005

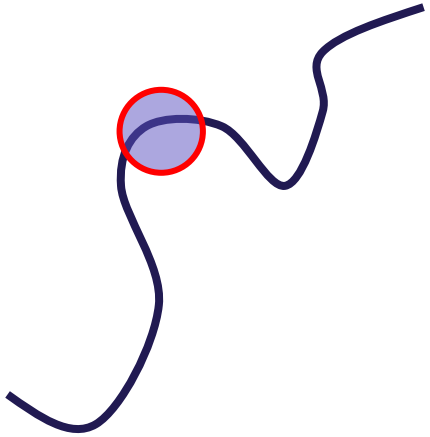
Robust feature descriptors

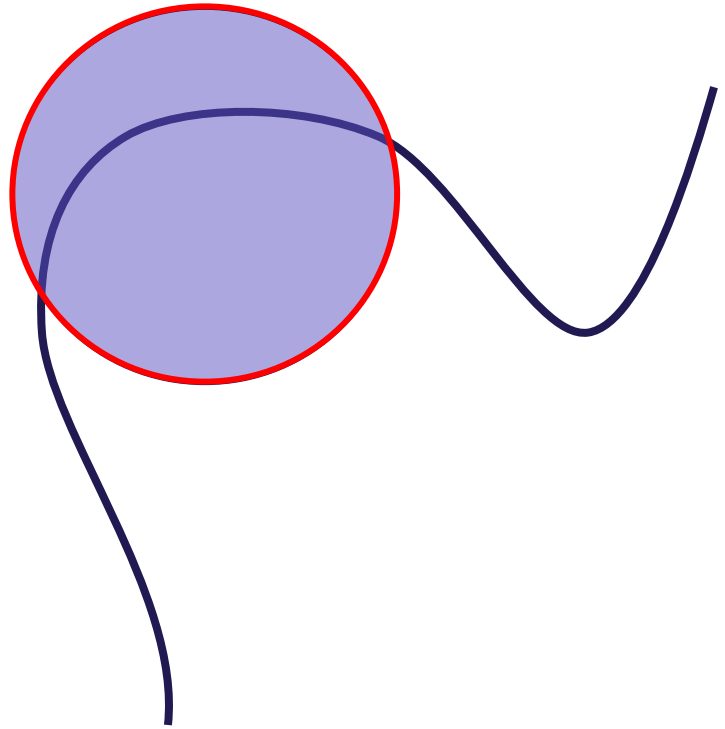
- Harris corners
  - Rotation invariant ☺
  - Not invariant to scale ☹
- HOG features
  - Lighting invariant ☺
  - Scale invariant – depending on detection window ☺
  - Not rotation invariant ☹
- We need a more general feature descriptor which is both rotation and scale invariant!

Scale invariant detection

- Consider regions (e.g. circles) of different sizes around a point

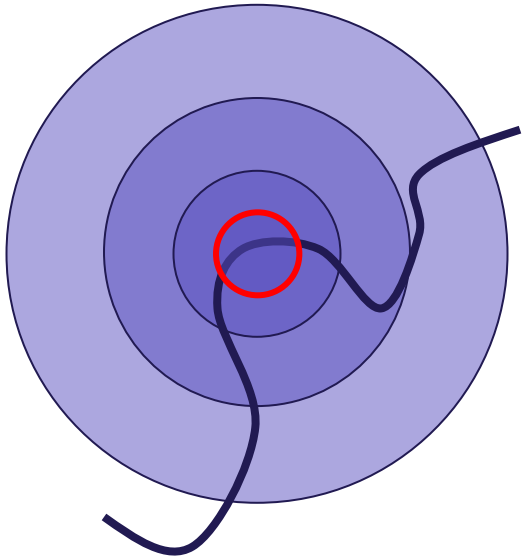


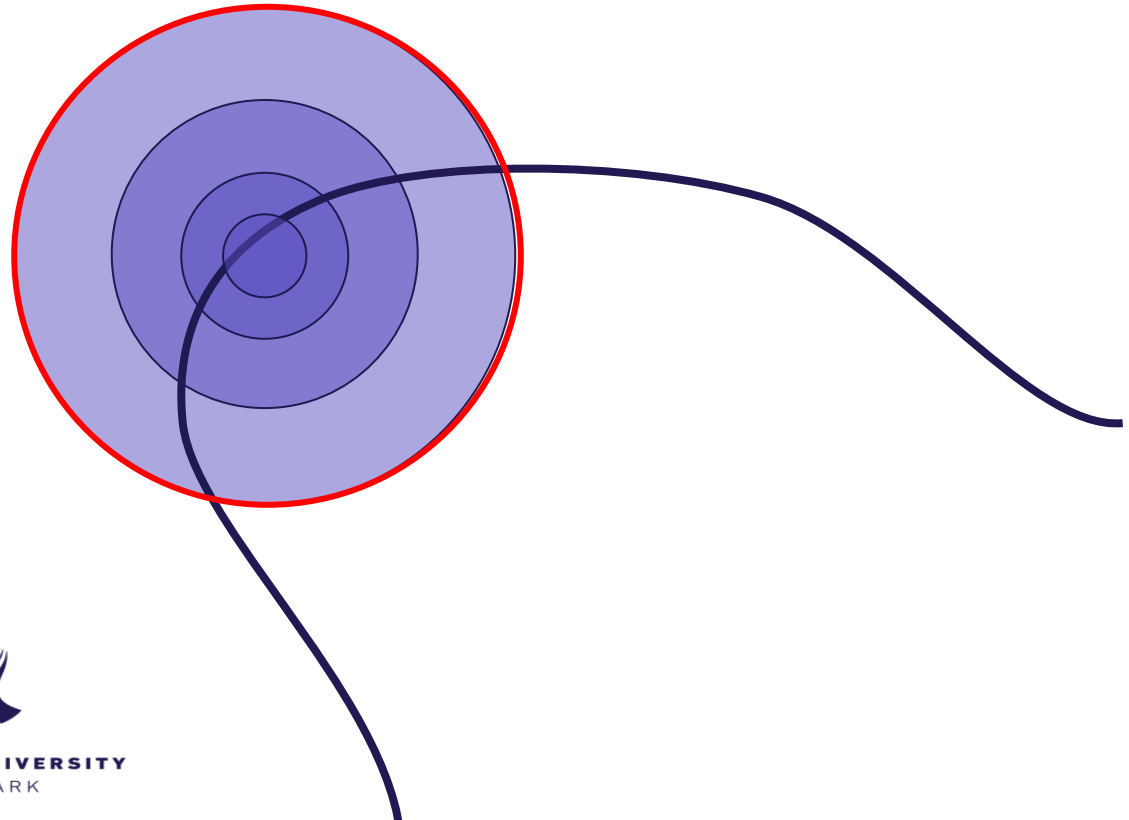




Scale invariant detection

- How do we choose corresponding sizes ***independently*** in each image?

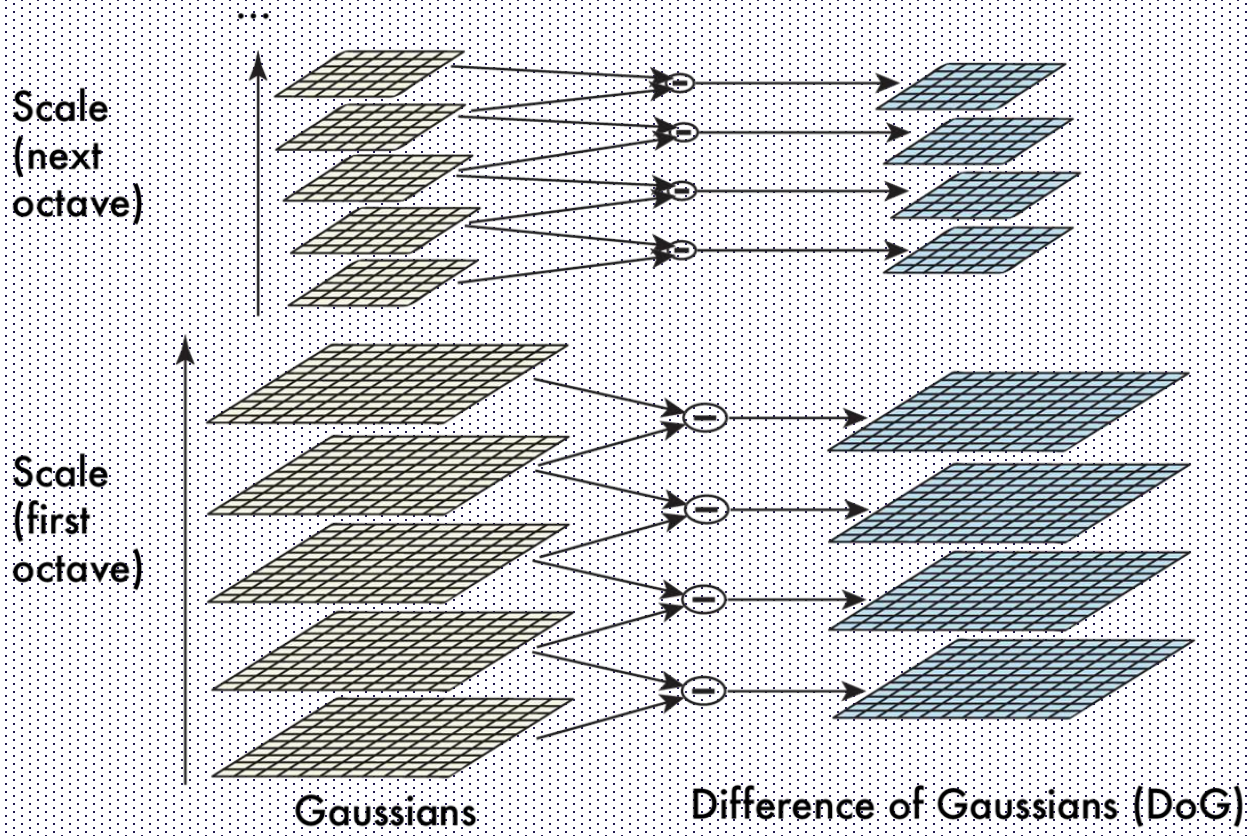




Scale Invariant Feature Transform (SIFT)

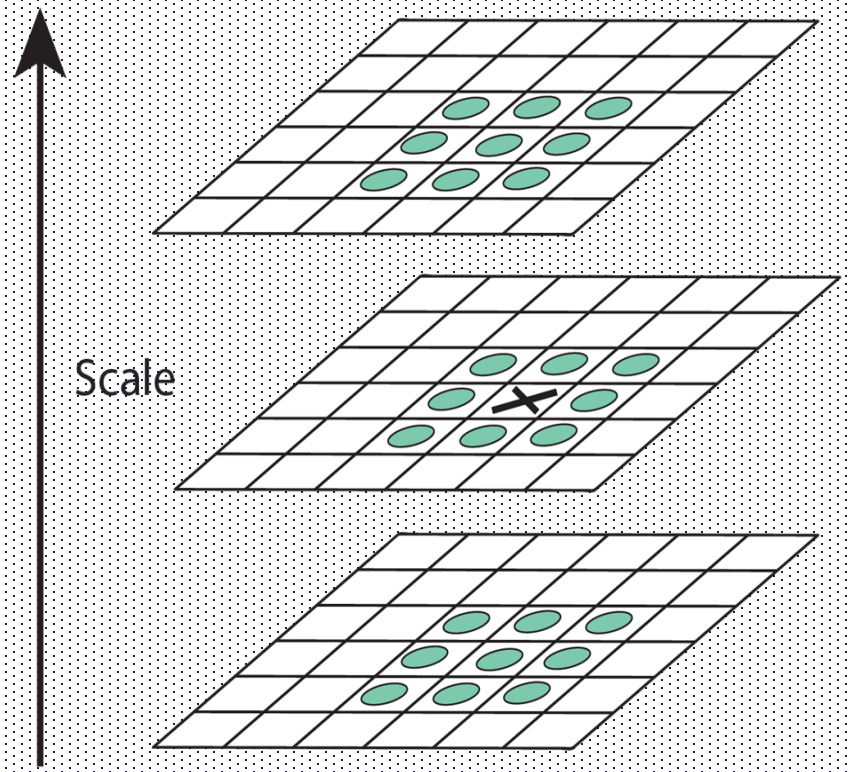
- Scale-space extrema detection
- Keypoint localization
- Orientation assignment
- Keypoint descriptor

Lowe: ”Distinctive Image Features from Scale-Invariant Keypoints”. International Journal of Computer Vision 2004



SIFT: extrema detection

- Extract DoG features at multiple scales



SIFT: extrema detection

- Find local extrema in location and scale
- Remove low contrast features and edges

SIFT: orientation assignment

Find main orientation of patches

- Orientation histogram of 36 bins
- Weighted by gradient magnitude and Gaussian-weighted circular window relative to scale
- Look for dominant directions
  - Any gradient within 80% of peak gets its own descriptor

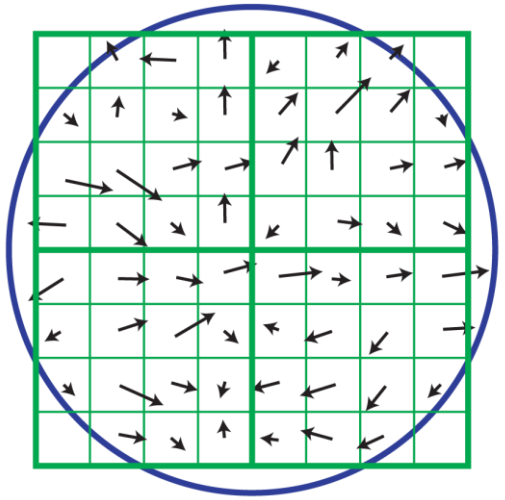
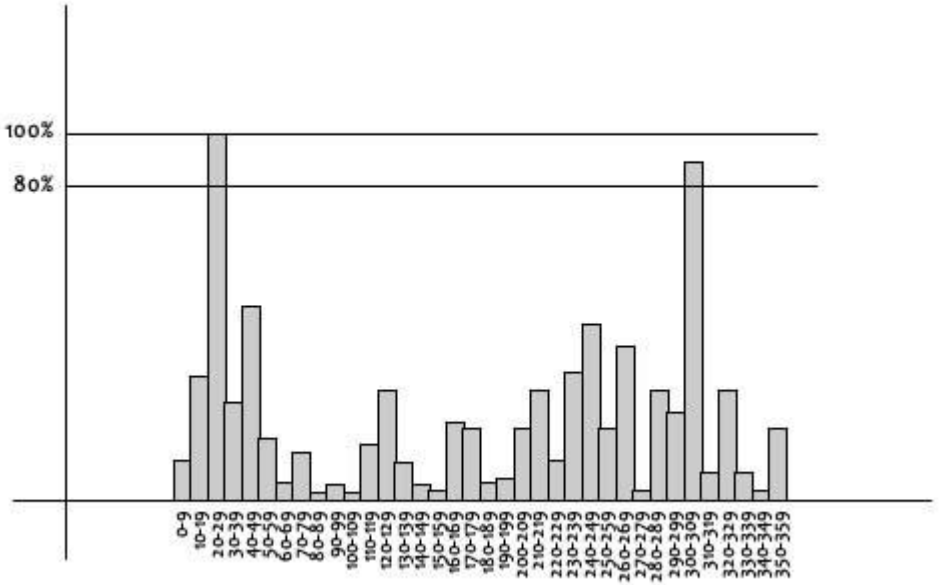


Image gradients



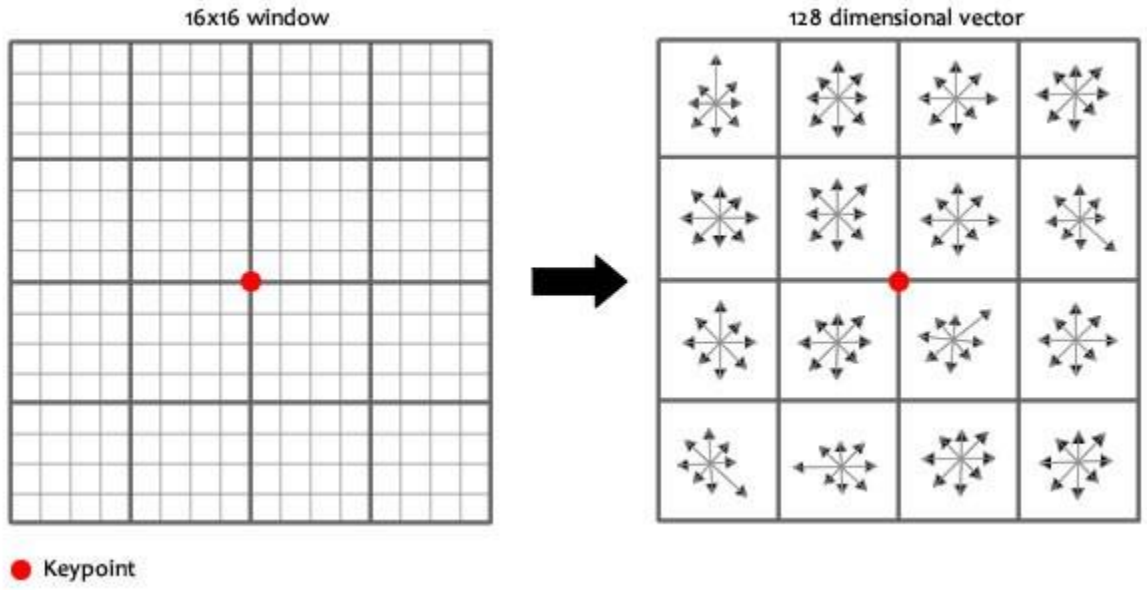
SIFT: keypoint descriptor

Feature descriptor

- Compute gradient directions and magnitudes around keypoint
  - Weighted by Gaussian window
  - Accumulate into orientation histograms (here 4x4 subregions)

- Rotation invariance:
  - Rotate relative to the keypoint orientation

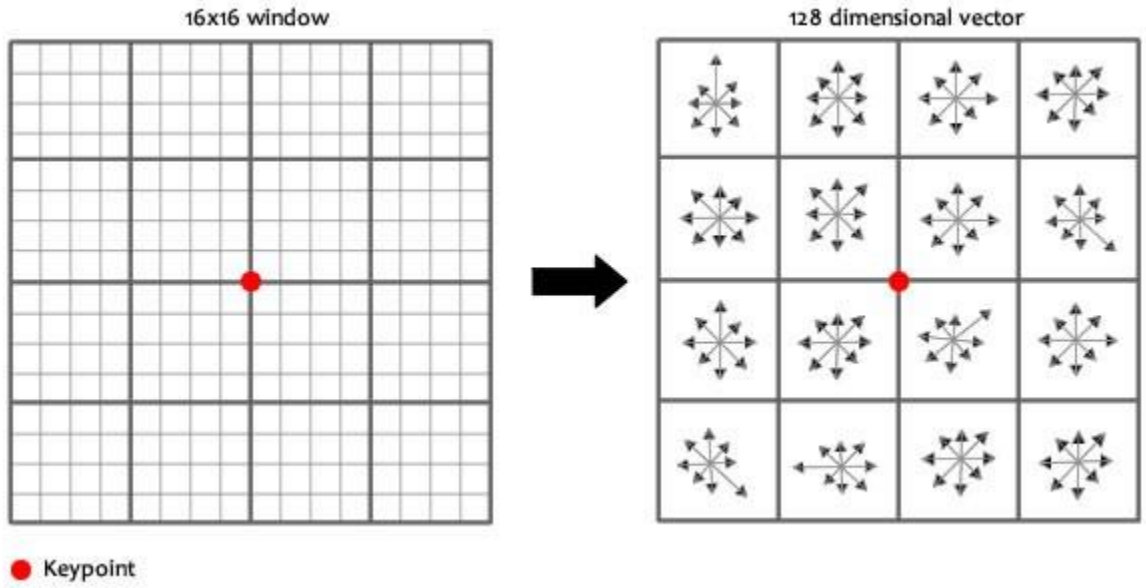
- Experimental values:
  - 4x4 array of histograms with 8 bins
  - 128 element feature vectors



SIFT: keypoint descriptor

Feature descriptor

- Illumination invariance:
  - Normalize vector to unit length
  - Threshold feature values at 0.2
  - Normalize to unit length again



SIFT - usage

- Find good keypoint, describe them
- Search for same keypoint to do detection, recognition, stitching, etc.

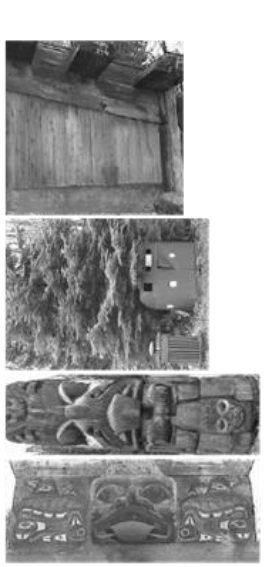




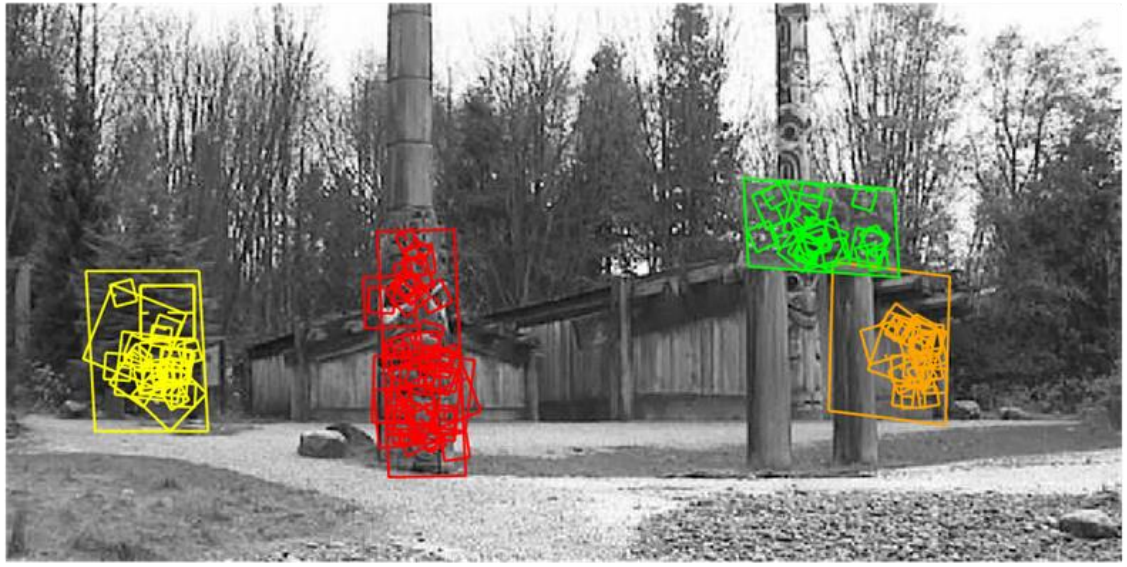




SIFT - usage







Robust features

SIFT has been state-of-the-art feature detector for many years

- Robust perfomance but many computations

Alternative:
SURF: Speeded Up Robust Features

Both SIFT and SURF are/were patented

- SURF is found in the “non-free library” – SIFT is now in normal feature library
- SURF can not be used for commercial purposes

ORB: Oriented FAST and Rotated BRIEF

- Good and fast alternative for real-time applications



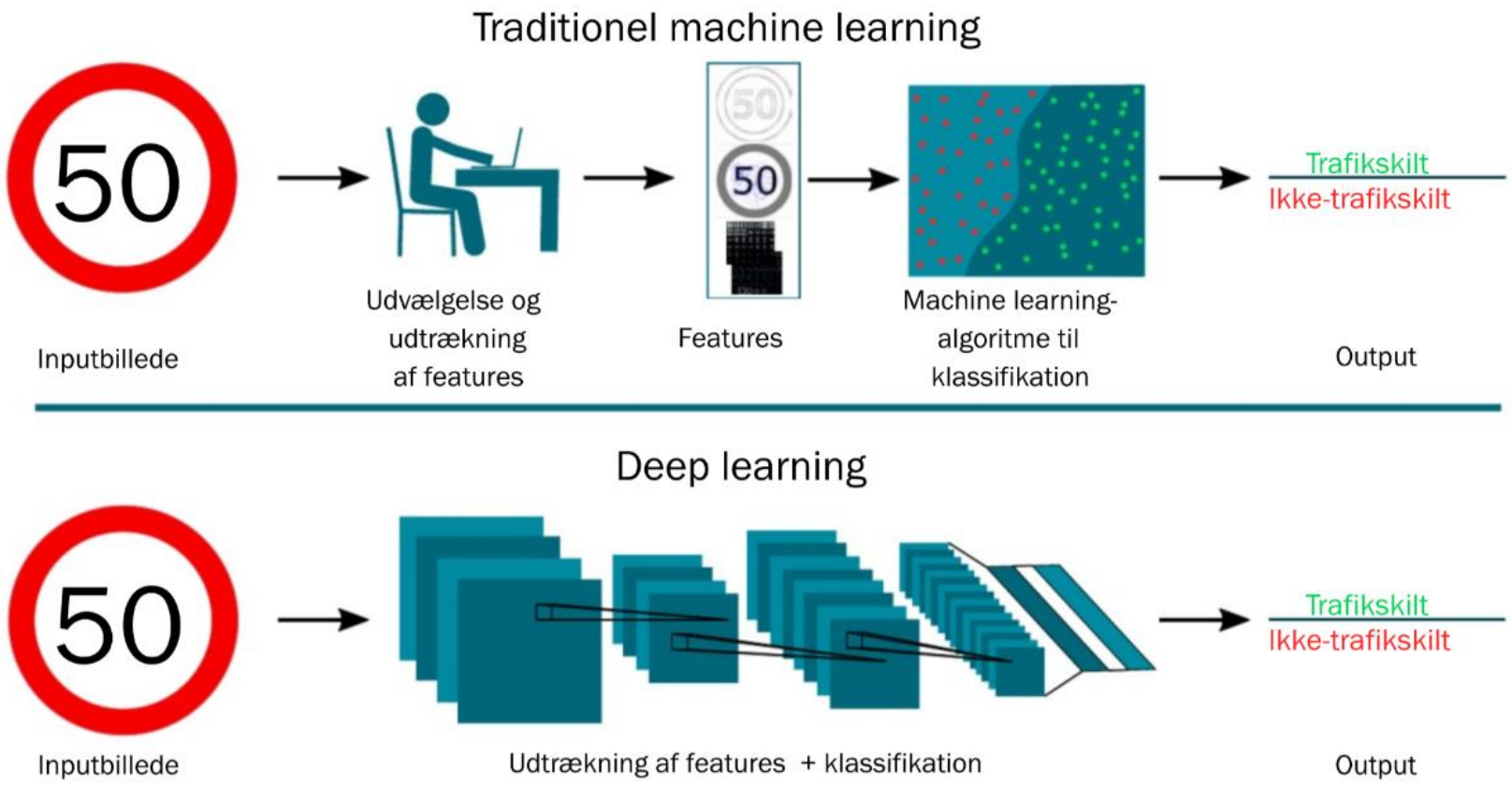
Feature detectors/descriptors

So many algorithms proposed, different pros and cons!

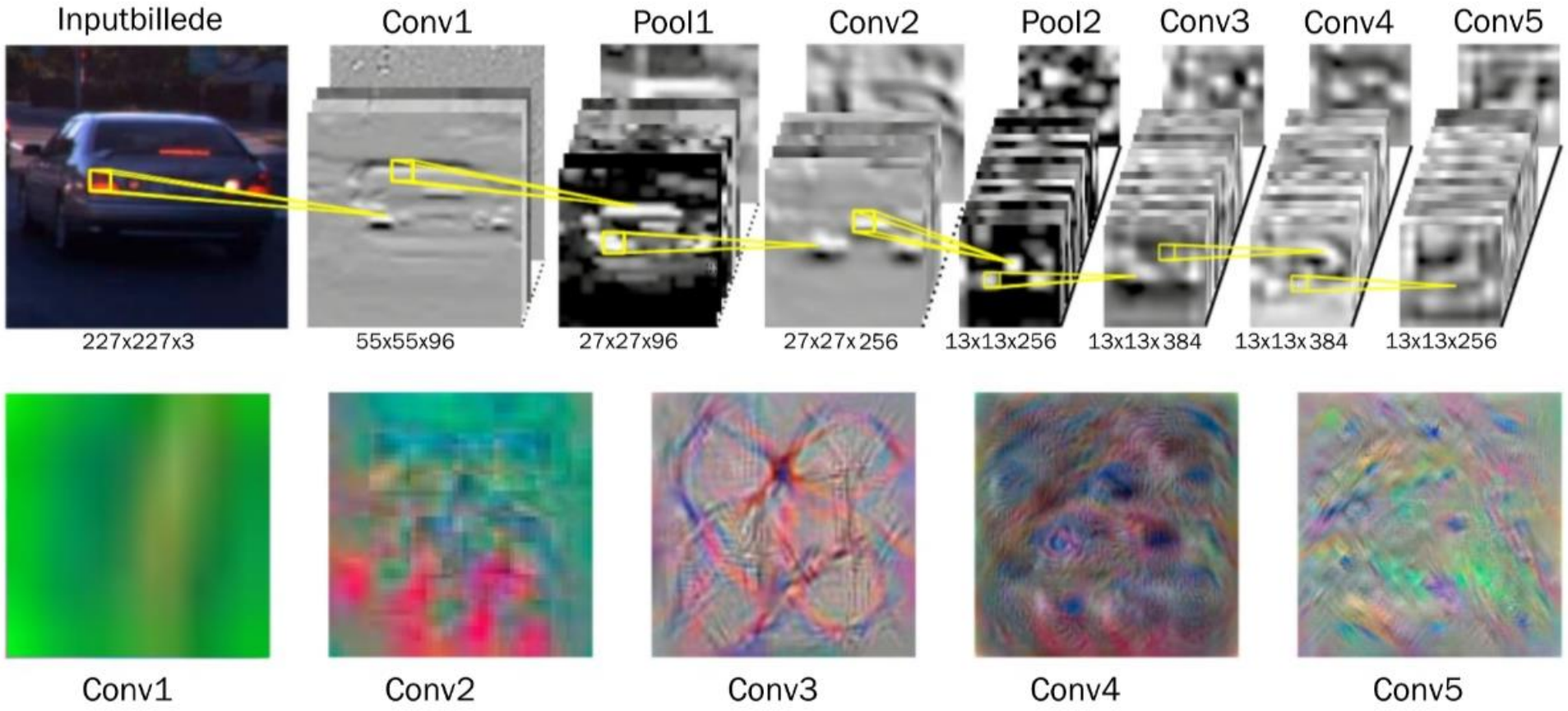
- SIFT
- SURF
- ORB
- BRIEF
- BRISK
- AKAZE
- FAST
- LATCH
- …



Deep features

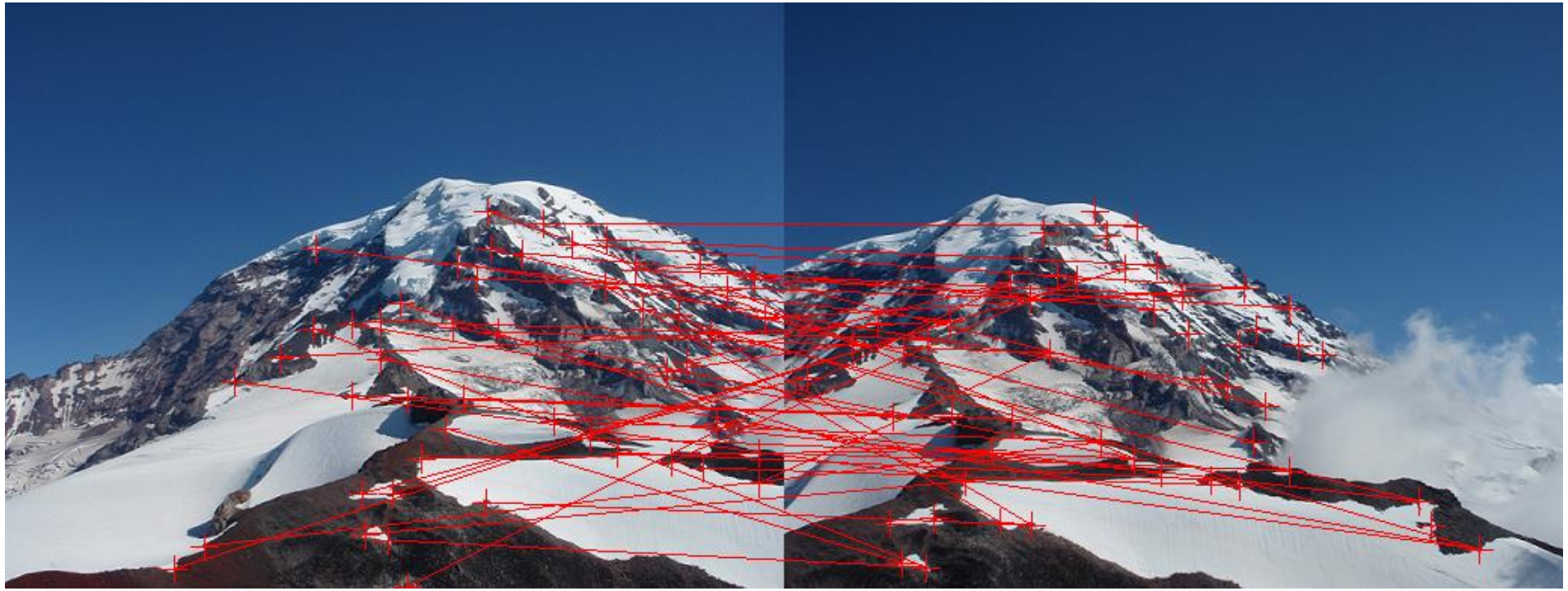


Deep features



Feature selection

- How should we calculate the transformation between images?
- Do matches agree??



Feature selection

Looking for a transformation model!

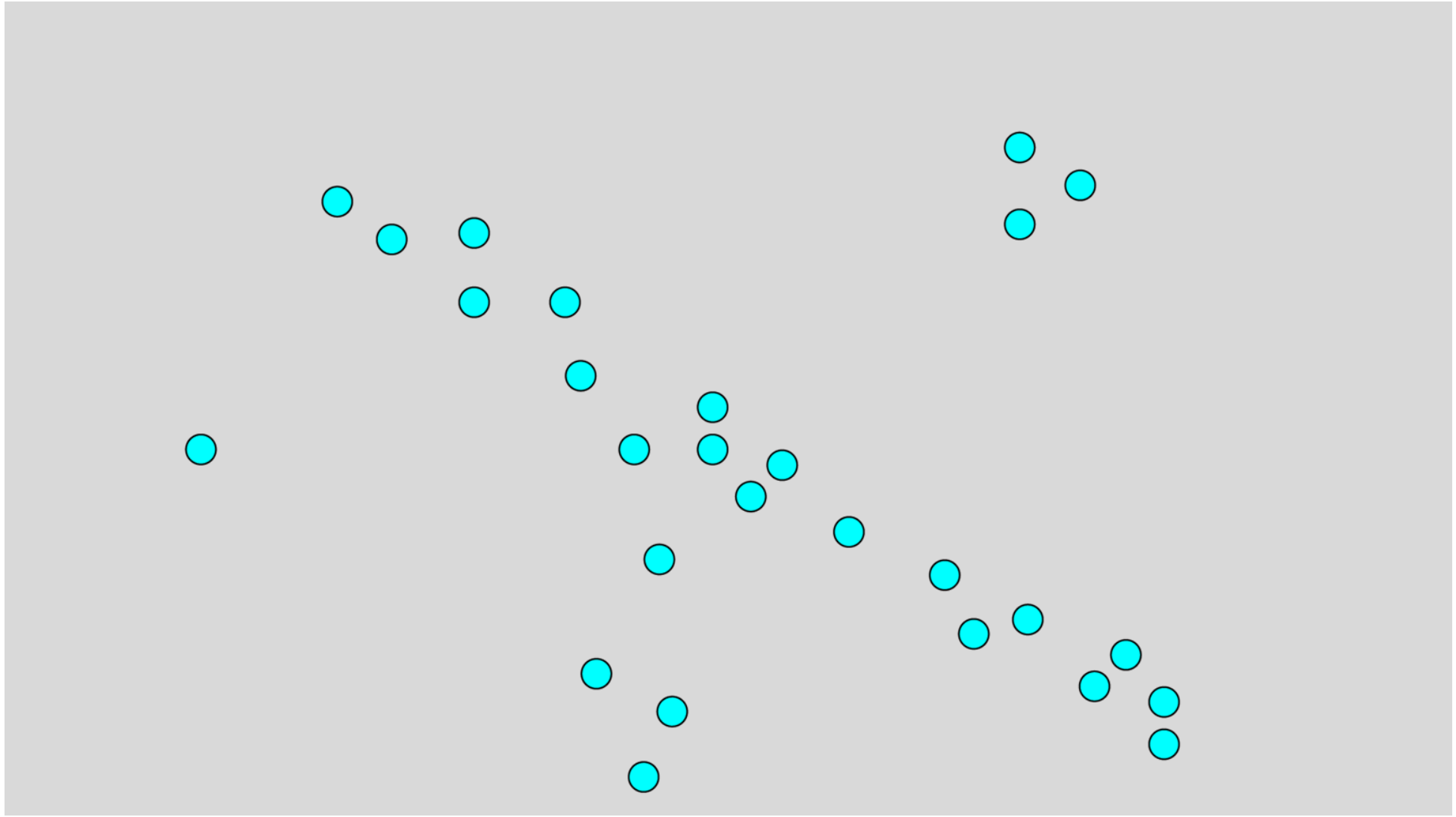
How can we fit model to inliers but ignore outliers?

- Least-squares method will find best solution, considering all points

- Try a bunch of models, see which ones are best
- Inliers will all agree on a model
- Outliers are basically random, will not agree

- RANSAC: RANdom Sample Consensus

RANSAC



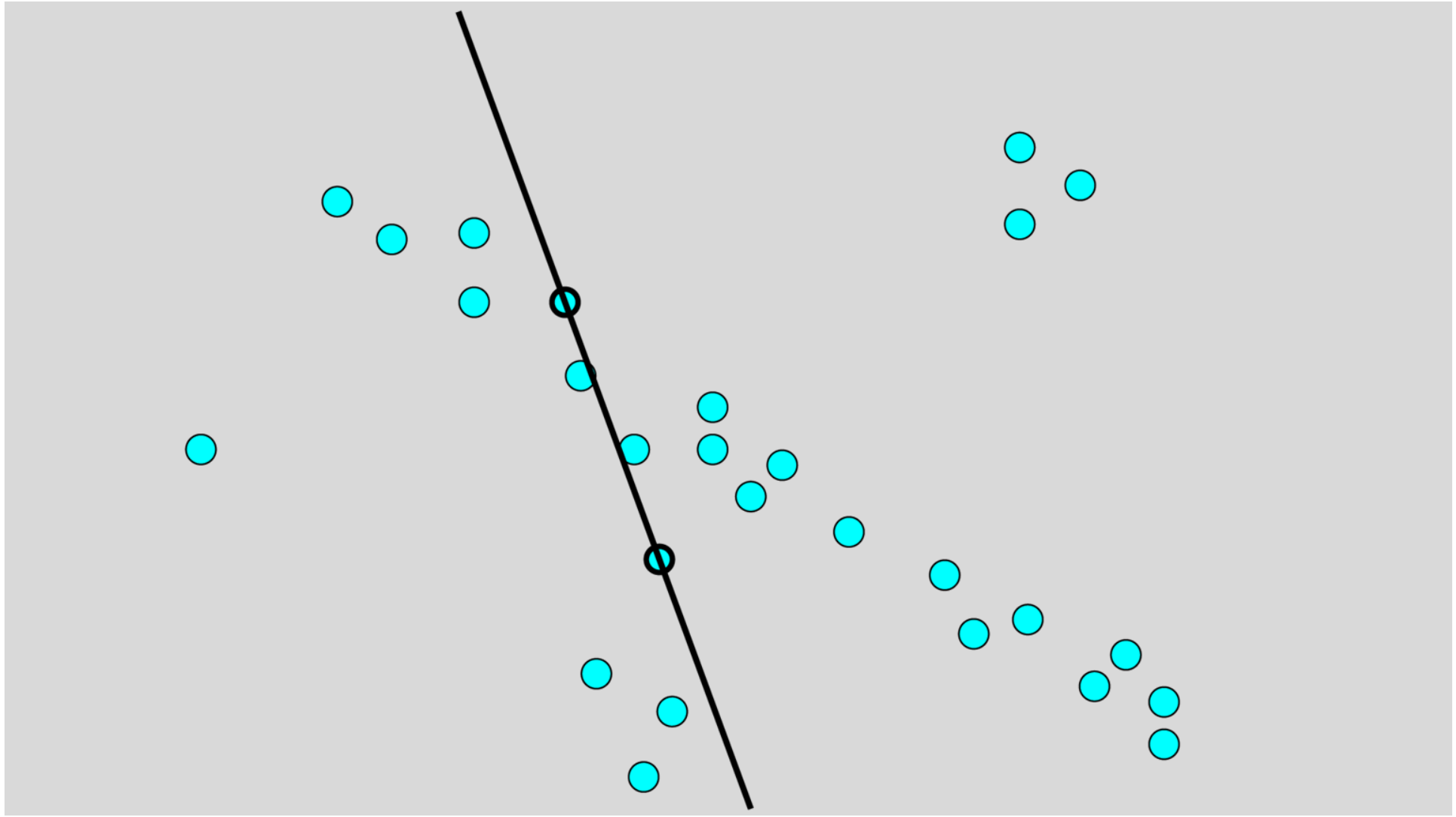
RANSAC

Procedure:

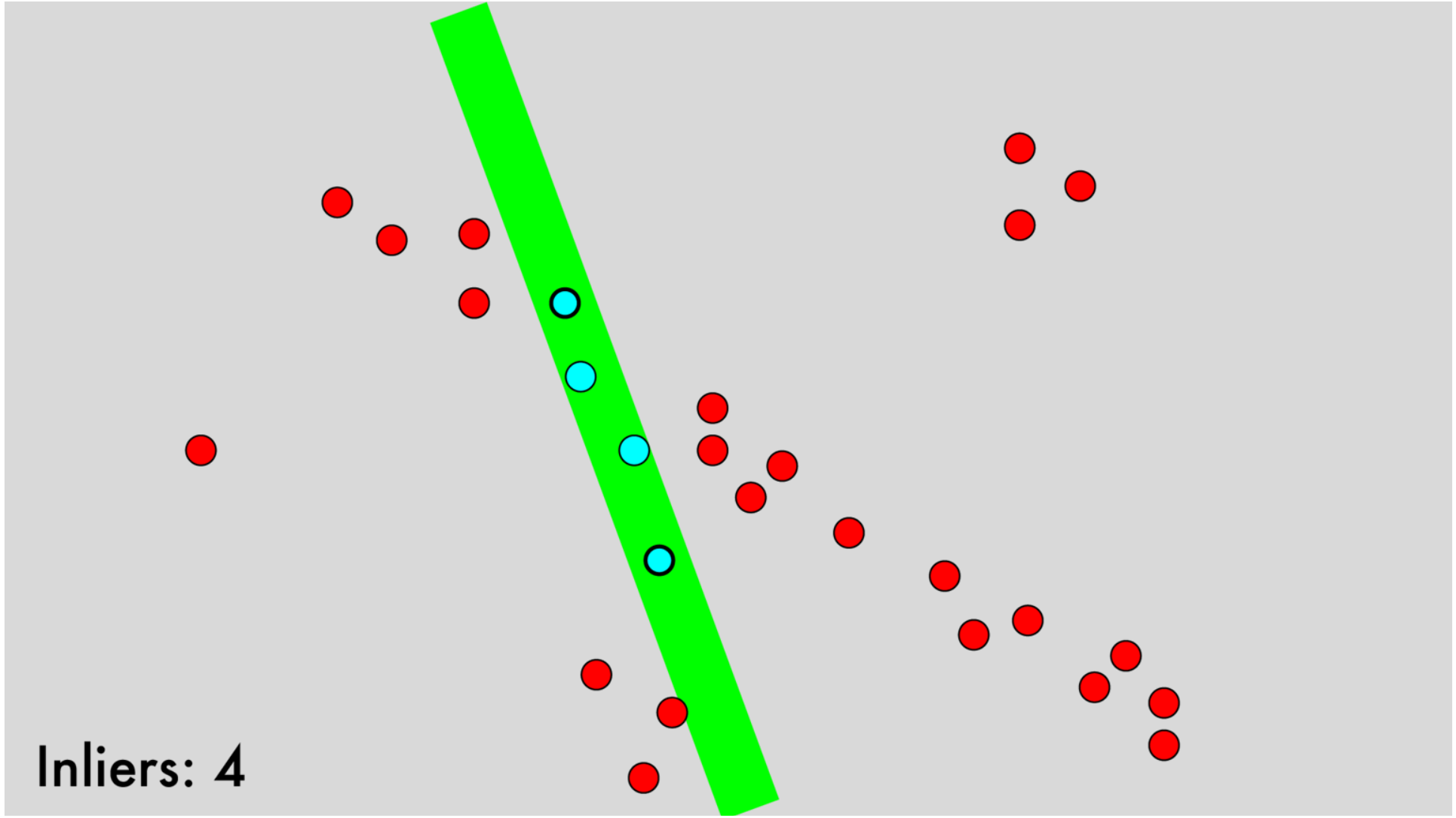
1. Select a random subset of the original data (size depends on model)
2. A model is fitted to this subset
3. Determine how many points from the set of all points fit with a predefined tolerance
4. If the fraction of the number of inliers over the total number points in the set exceeds a predefined threshold, re-estimate the model parameters using all the identified inliers and terminate
5. Otherwise, repeat steps 1 through 4 (maximum of N times)



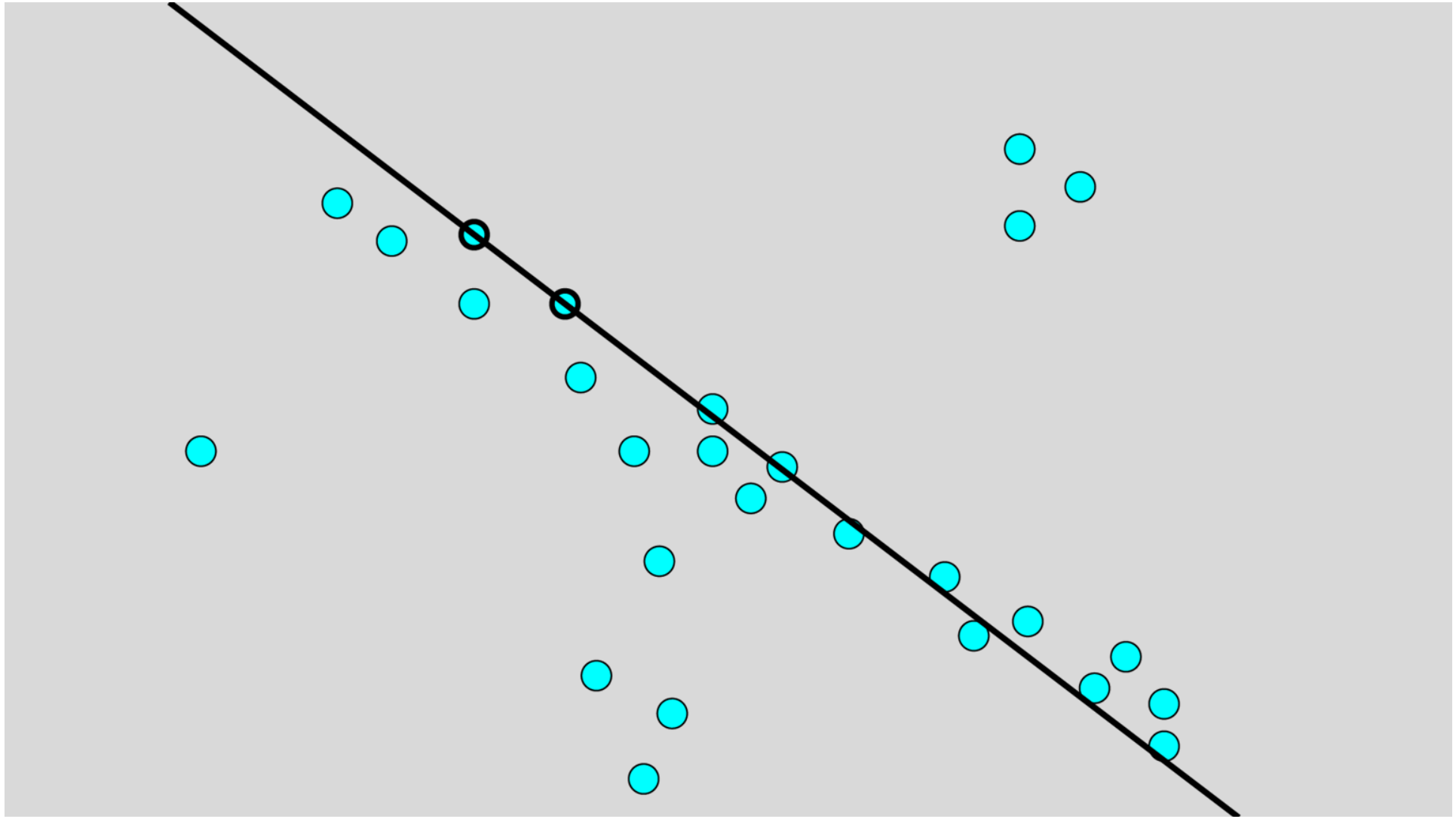
RANSAC



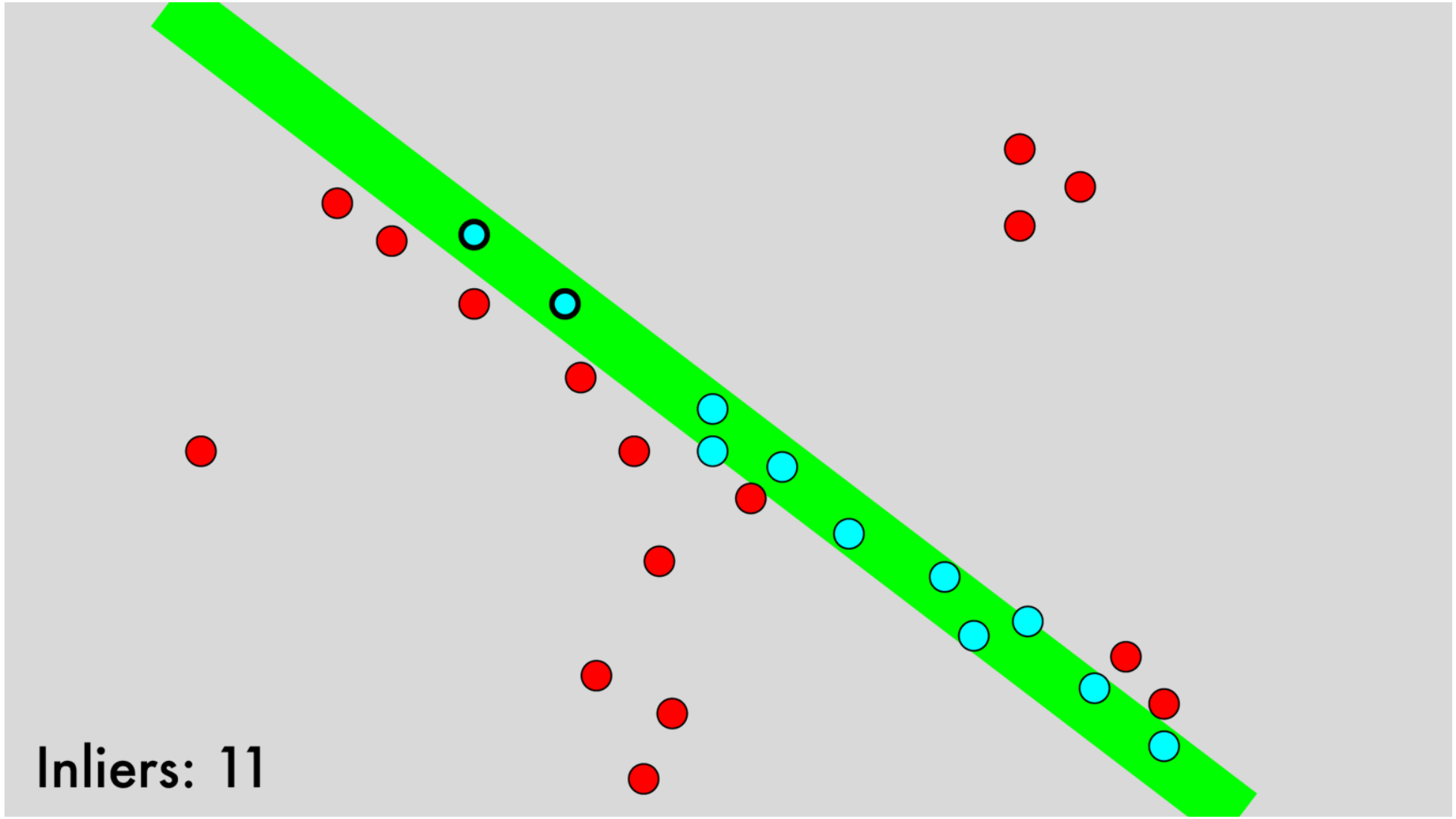
RANSAC



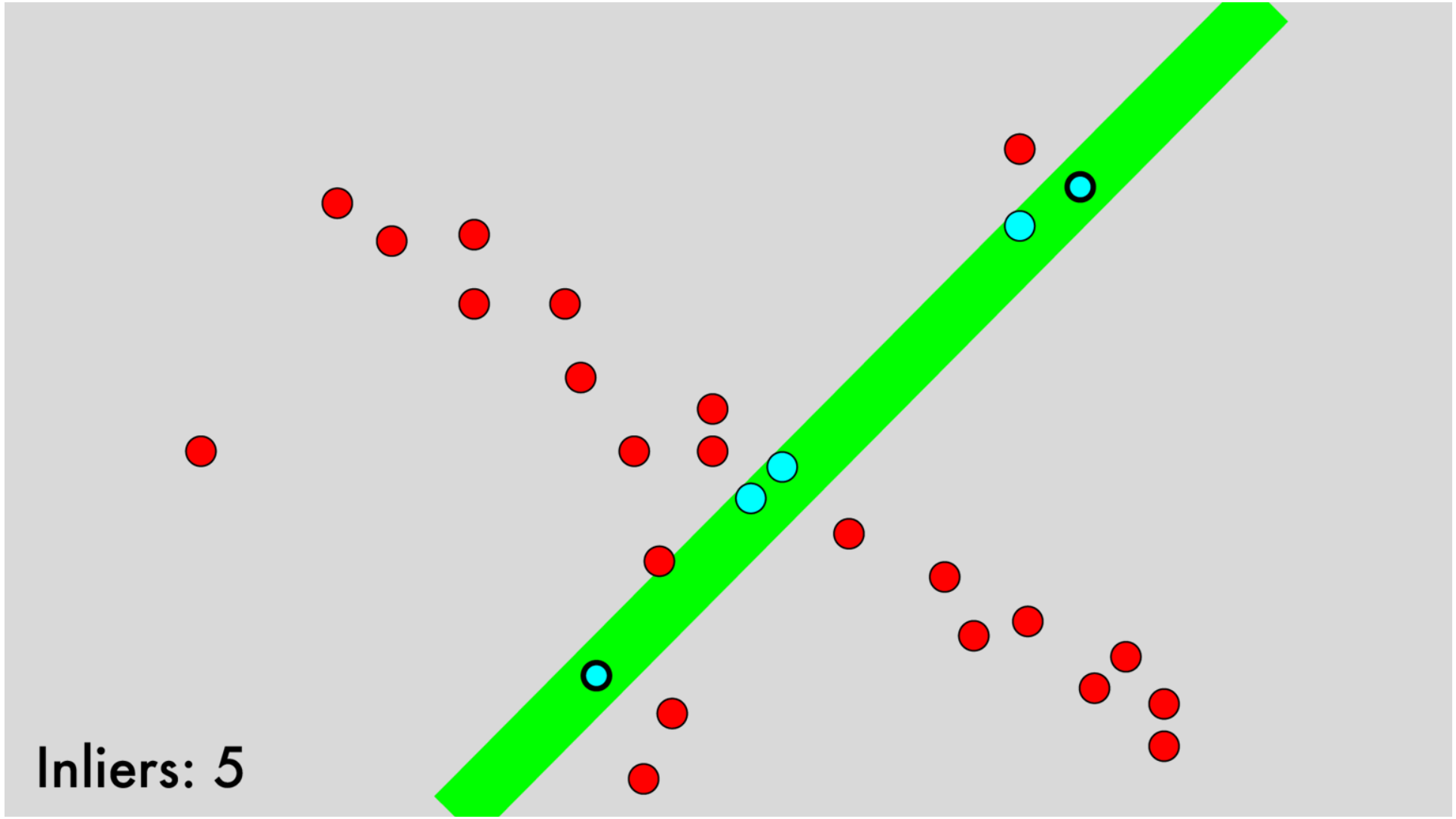
RANSAC



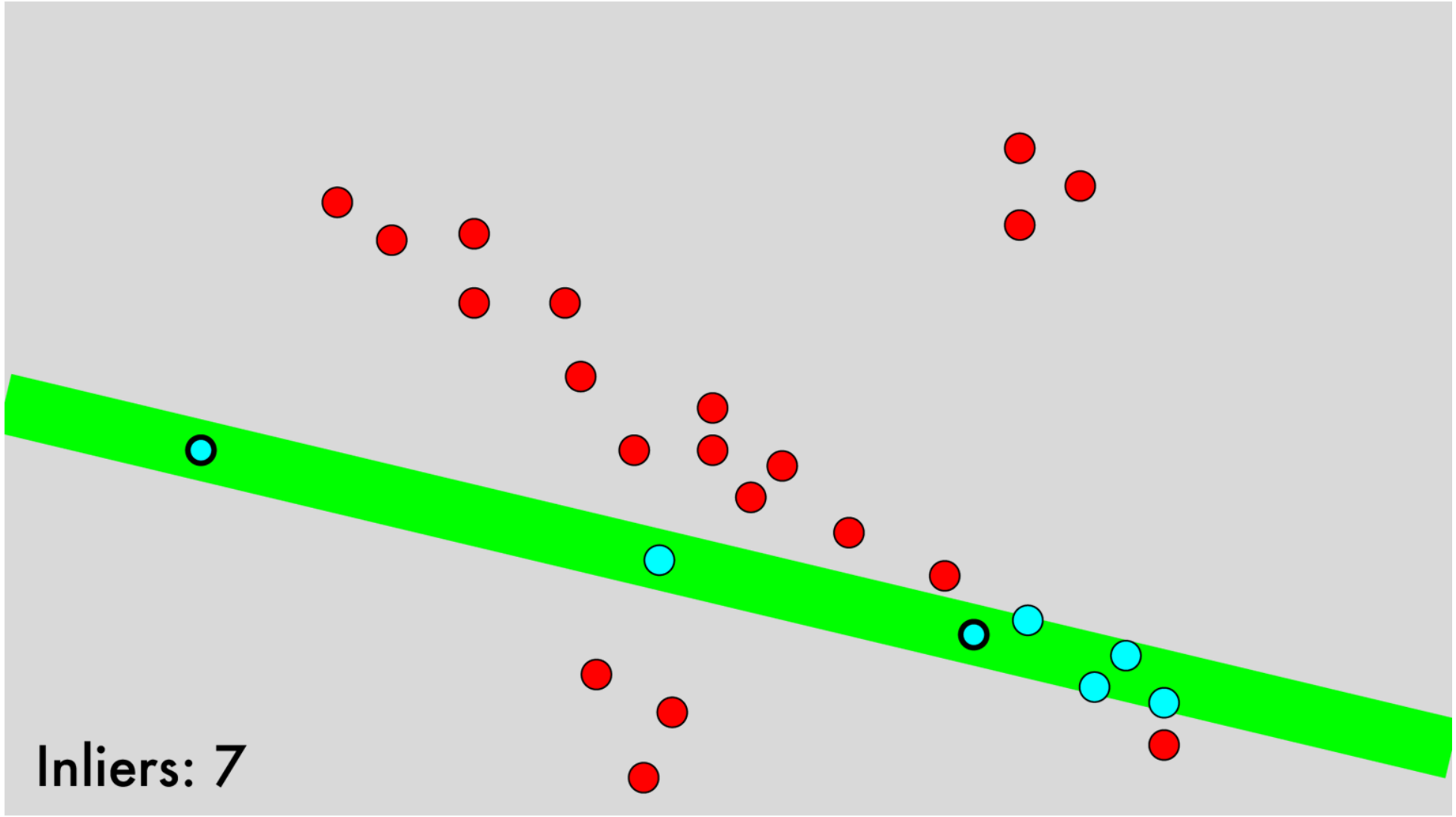
RANSAC



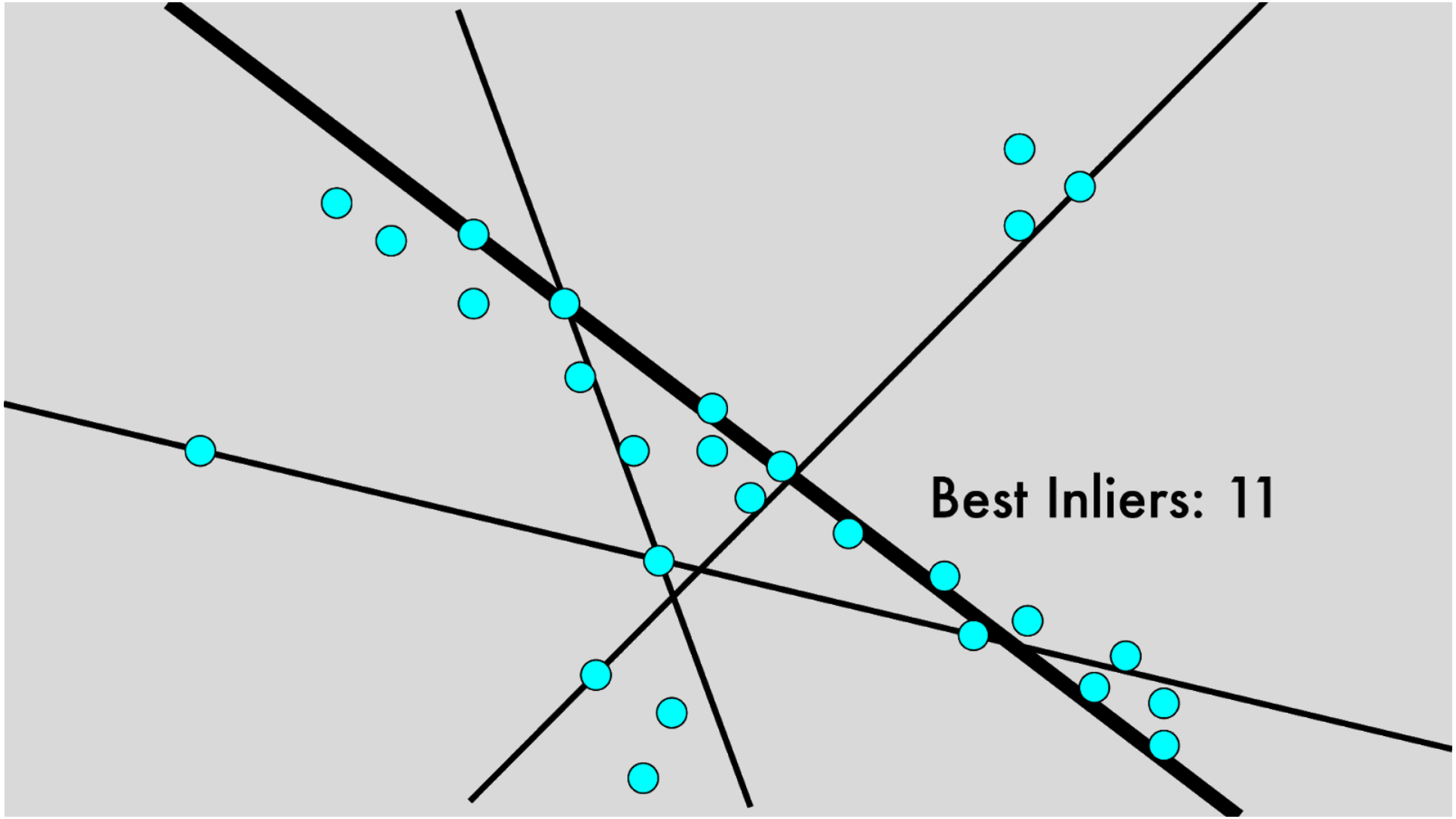
RANSAC



RANSAC



RANSAC



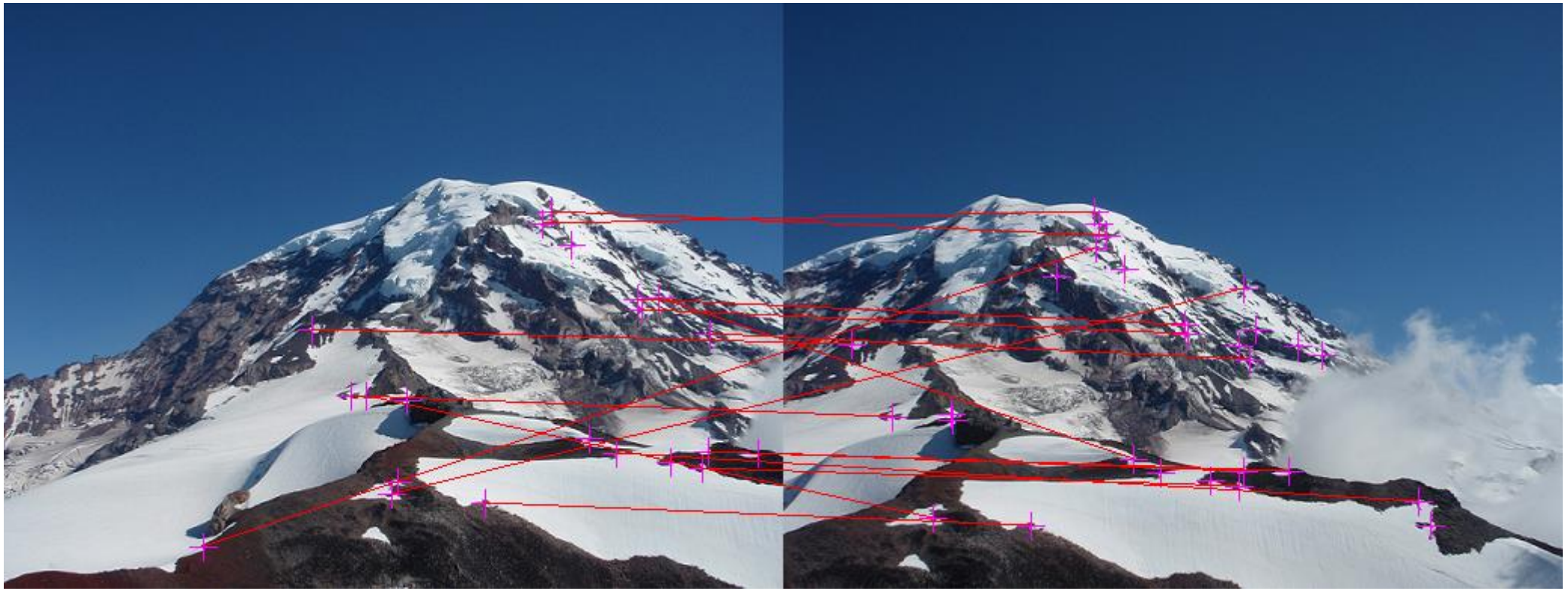
RANSAC

Parameters: *data*, *model*, *n* points to fit model, *k* max iterations, *t* threshold, *d* ”good” fit cutoff
Algorithm:

```
bestmodel = None
bestfit = 0
while i < k:
    sample = draw n random points from data
    Fit model to sample
    inliers = data within t of model
    if inliers > bestfit:
            Fit model to all inliers
            bestfit = fit
            bestmodel = model
            if inliers > d:
                    return model
return bestmodel
```

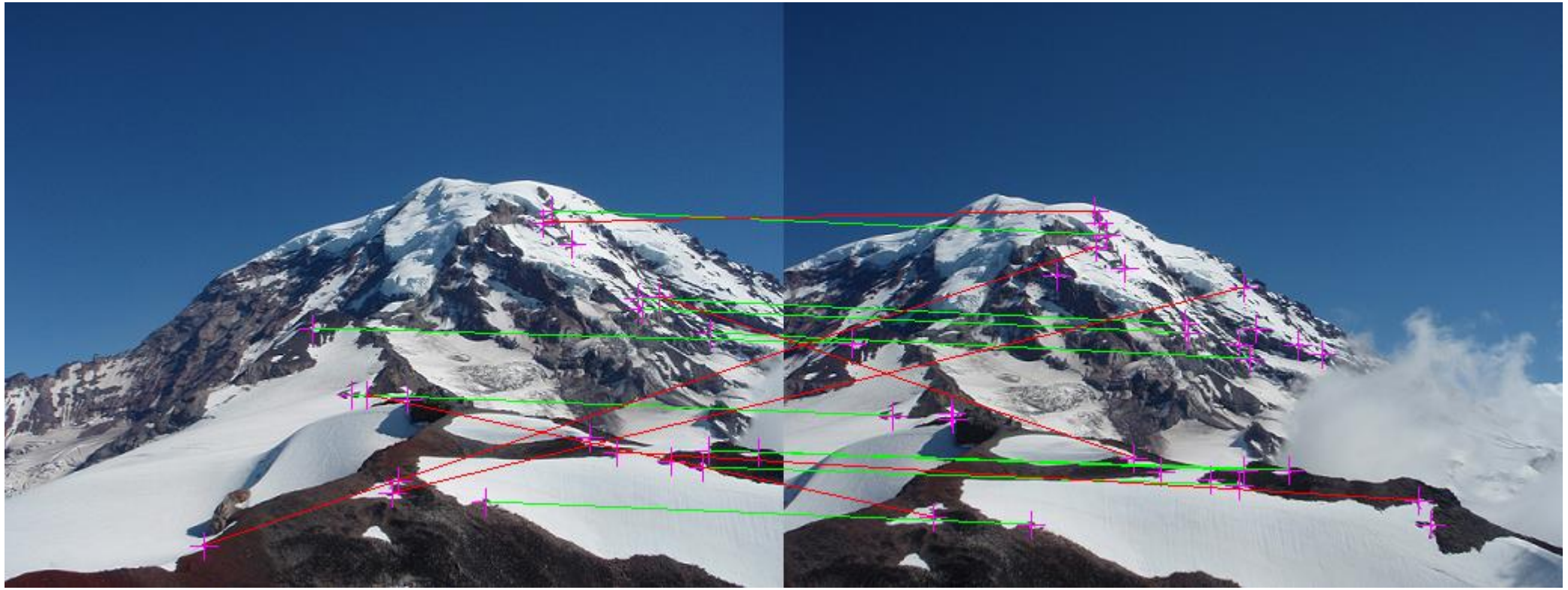
RANSAC

- Works well even with extreme noise



RANSAC

- Works well even with extreme noise



RANSAC

Parameters: *data*, *model*, *n* points to fit model, *k* iterations, *t* threshold, *d* ”good” fit cutoff

- Lots of tunable parameters
- Want high probability of recovering “right” model

- t: often quite small, assume “good” inliers
- n: should be just enough to fit model, no extra
- k: can be very high
- d: should be >> n



Dimensionality reduction

- Goal: To reduce the number of dimensions (of a feature vector) and still preserve the variance of data

- Methods presented today:
  - Principal Component Analysis (PCA)
  - Linear Discriminant Analysis (LDA)

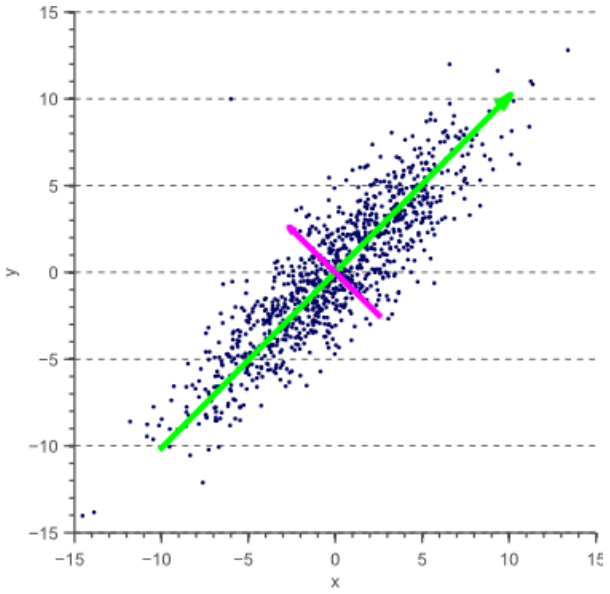
- Other methods:
  - ISOMAP
  - Multidimensional scaling
  - …

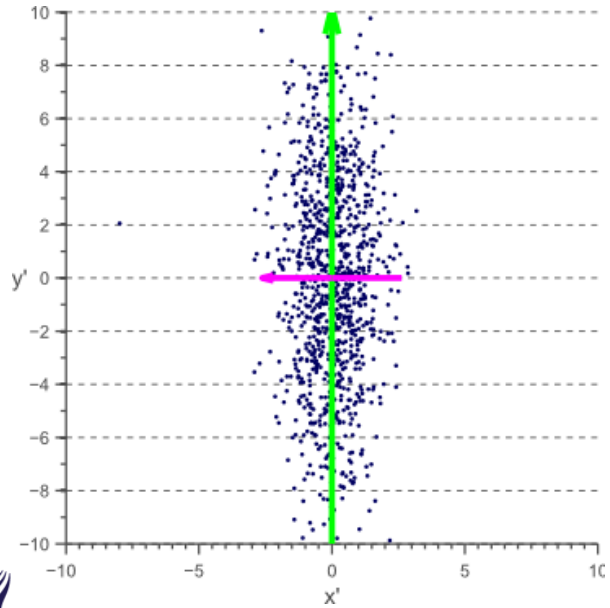
Principal Component Analysis

- Based on the covariance matrix

$$\Sigma = \begin{bmatrix} 16.87 & 14.94 \\ 14.94 & 17.27 \end{bmatrix}$$

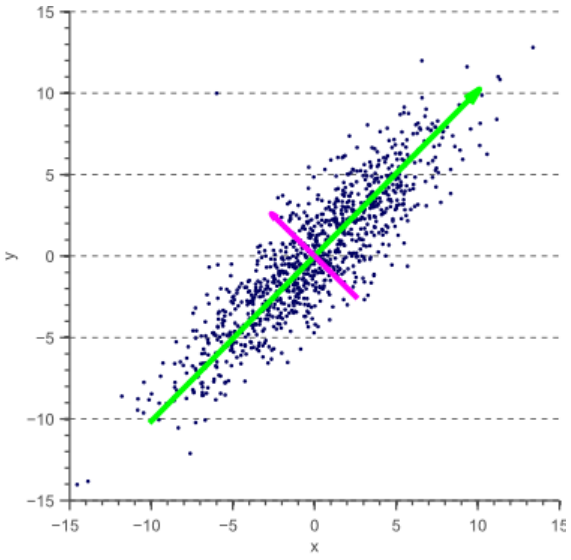
$$\Sigma' = \begin{bmatrix} 1.06 & 0.0 \\ 0.0 & 16.0 \end{bmatrix}$$

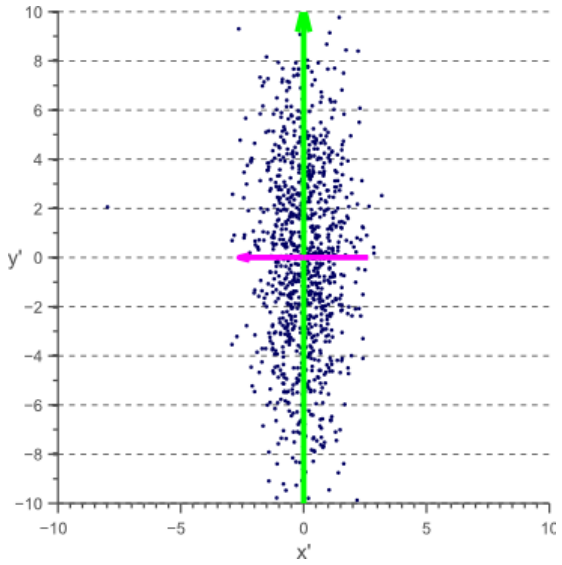


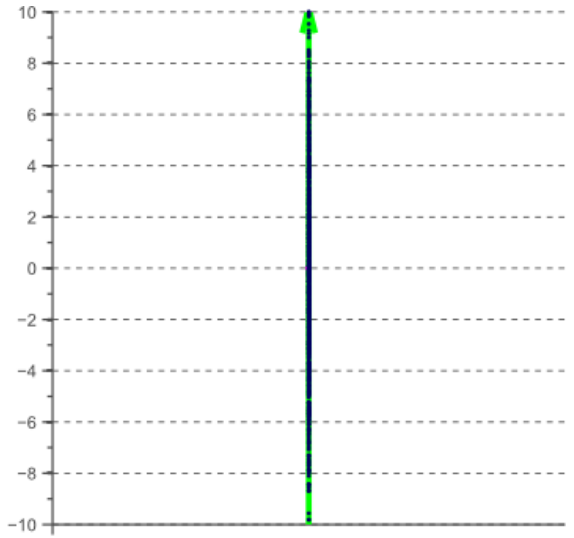


Principal Component Analysis

- Keeping only the dimension(s) with highest variance

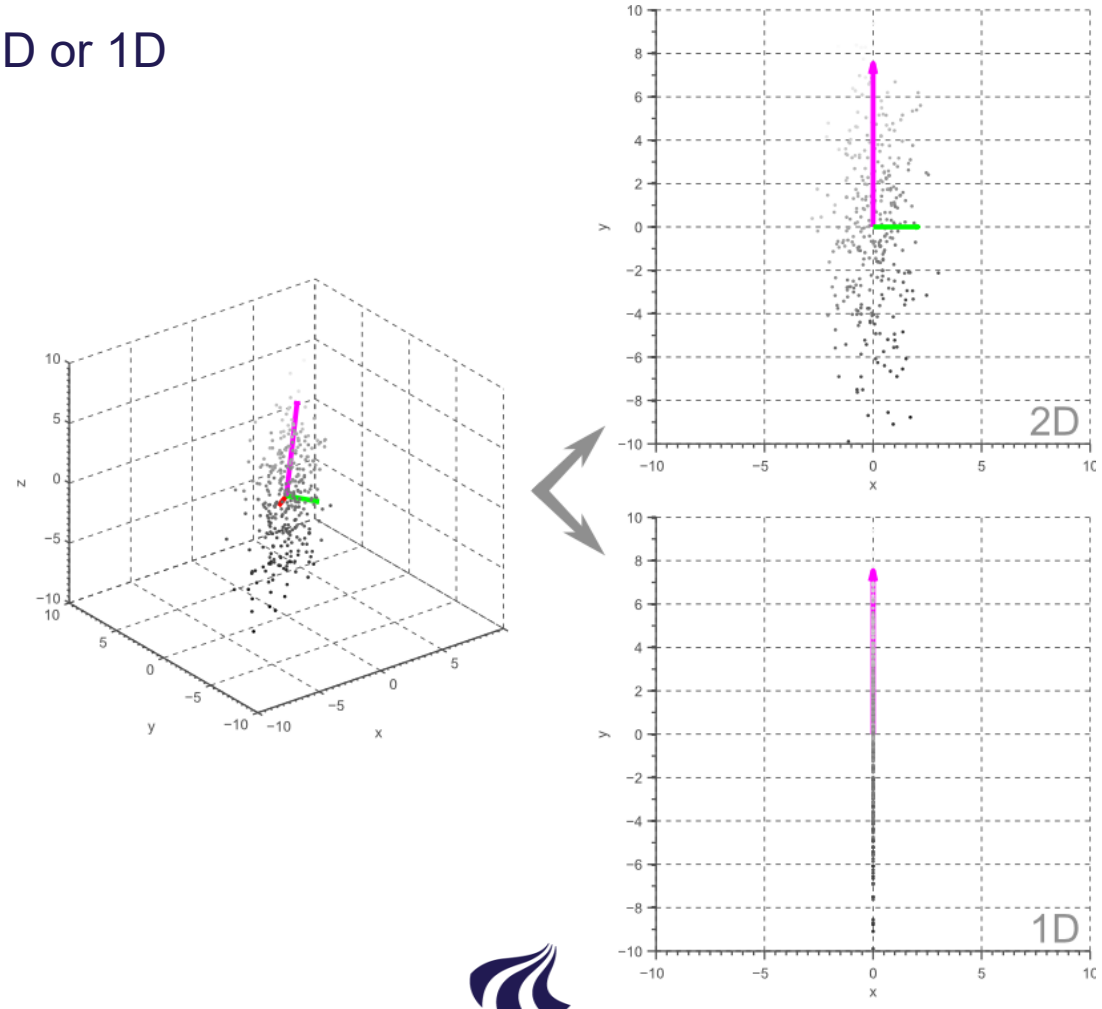






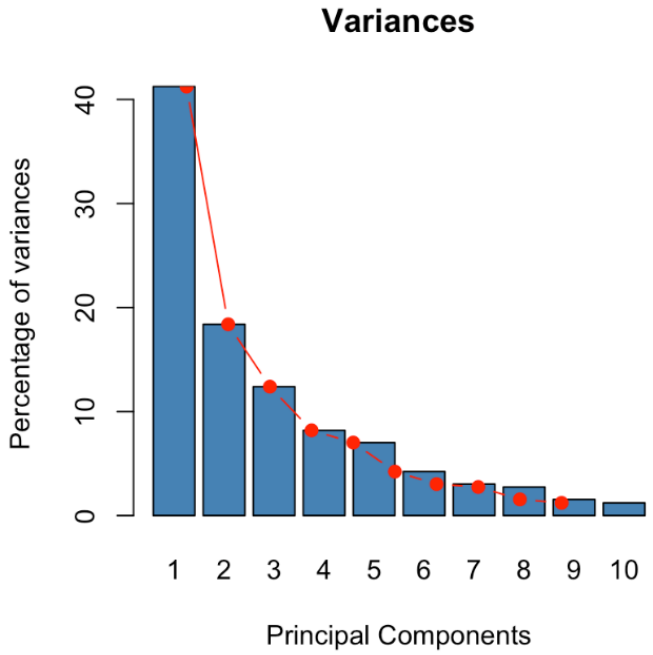
Principal Component Analysis

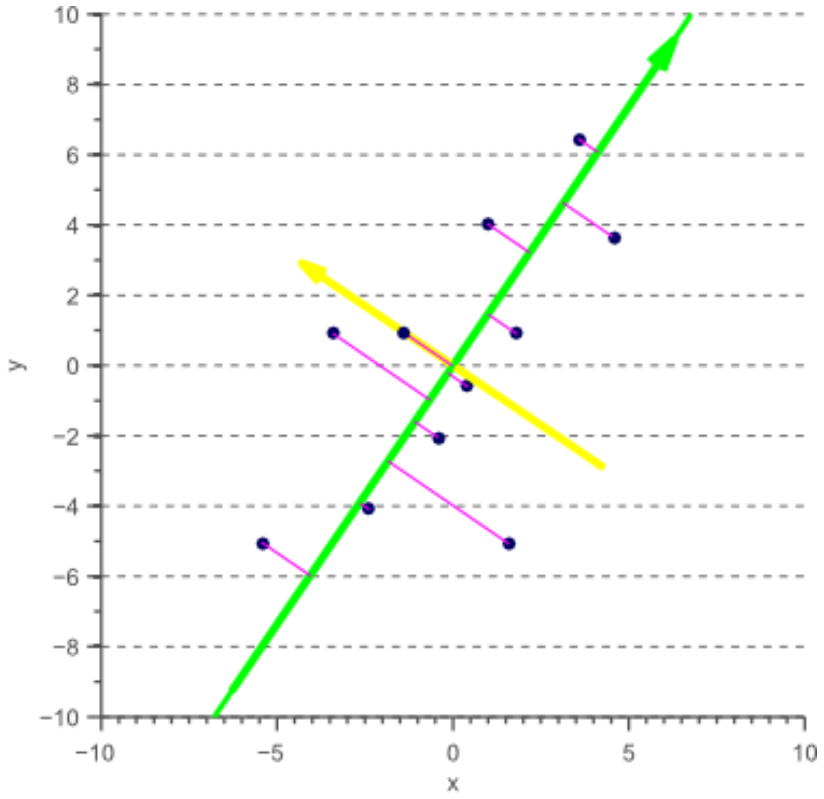
- Example: 3D reduced to 2D or 1D



Principal Component Analysis

- Find eigenvectors and corresponding eigenvalues
- Variance explained by each eigenvector can be found from eigenvalues





Principal Component Analysis

Method:

1. Standardize data
    - Calculate mean and subtract it to center the data around 0.
2. Calculate covariance matrix of standardized data
3. Find eigenvalues and corresponding eigenvectors
4. Transform data using *n* eigenvectors with largest eigenvalues
    - *n* decides the dimensionality of the transformed data



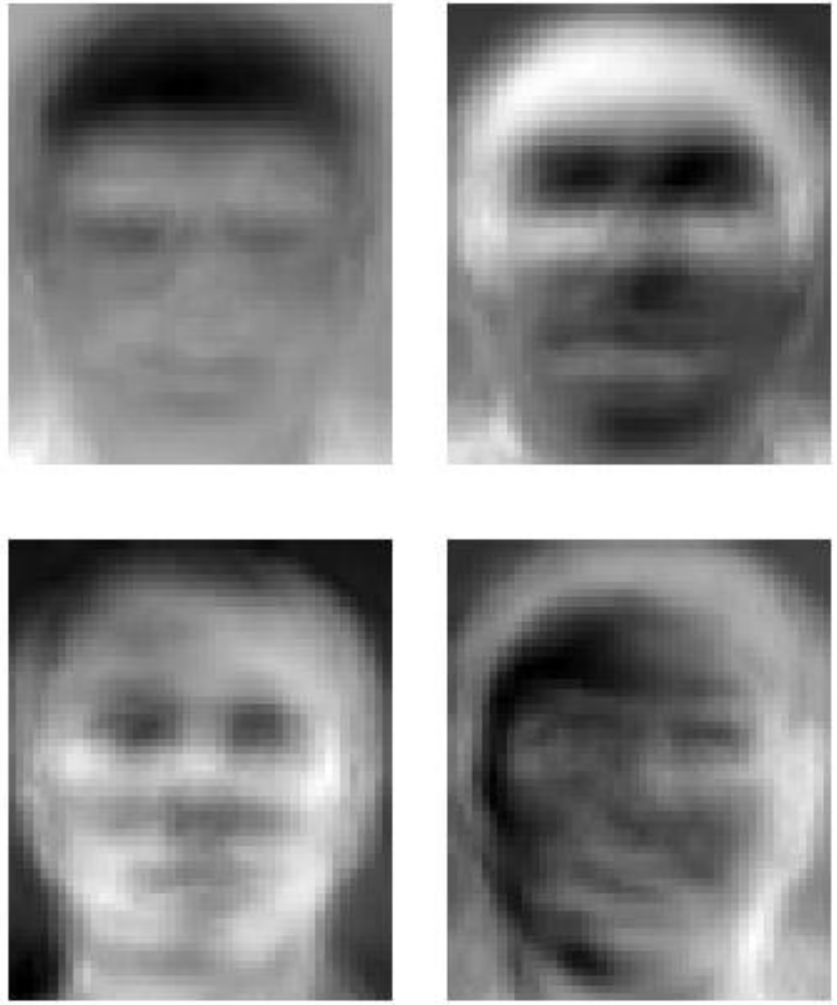
Principal Component Analysis

- Application: Early days face recognition
  - Eigenfaces
- Approach: Treating each pixel in an image as a feature
- Problem: Small image of just 32x32 pixels = 1024 features!
- Solution: Apply PCA to reduce dimensionality

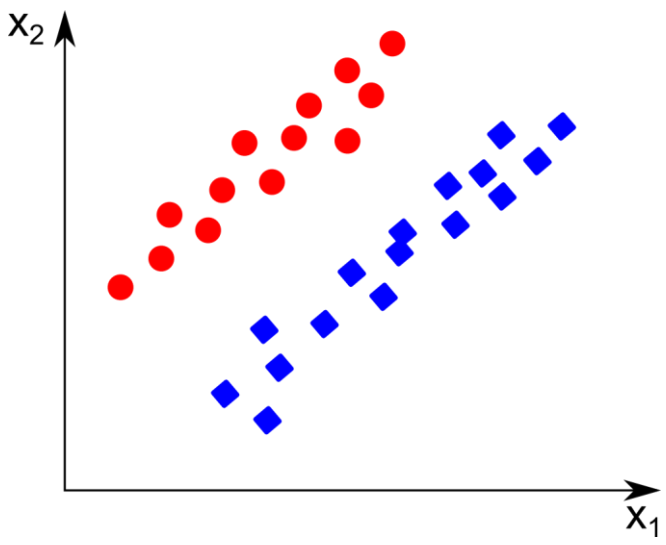


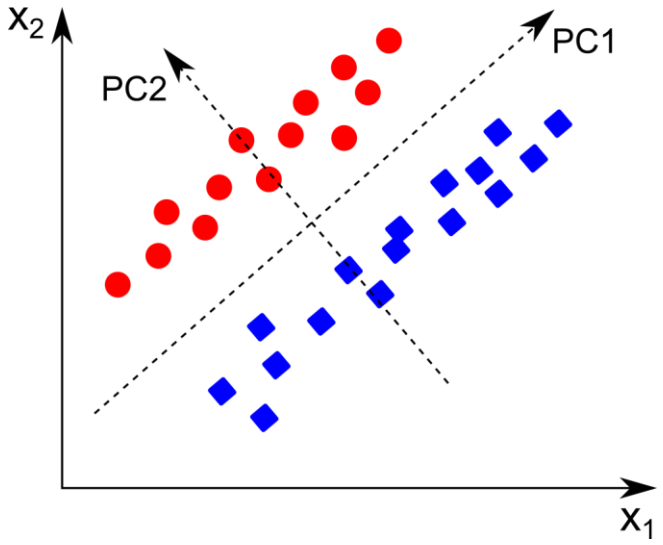
Principal Component Analysis

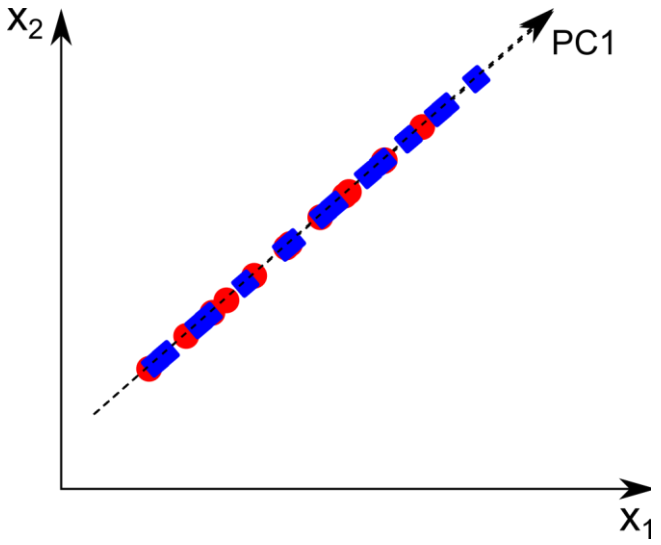
- Eigenfaces
- For n-dimensional data each eigenvector is also n-dimensional
- E.g.: 32x32 pixel images = 1024 dimensions
- Vector can be reshaped to 32x32 to visualize ”eigenfaces”



Principal Component Analysis





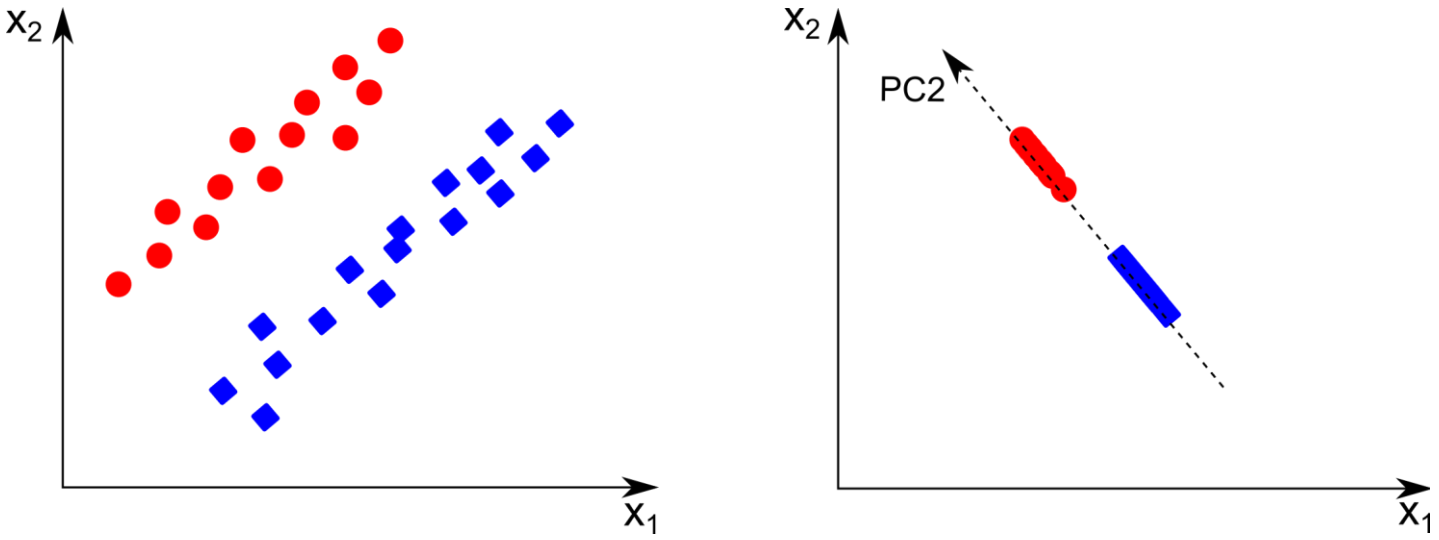


Limitation:

- PCA is unsupervised = uses unlabeled data
  - Seek the dimensions with largest variance between all data

Linear Discriminant Analysis

- Idea: With labelled data, seek the directions that are efficient for discrimination between classes



- The terms Linear Discriminant Analysis (LDA) and Fisher's Linear Discriminant are often used interchangeable

Linear Discriminant Analysis

- Find the projection of data such that the ratio of between-class scatter S_B and within-class scatter S_W is maximised:

$$W_{opt} = arg \max_W \frac{|W^T S_B W|}{|W^T S_W W|}$$

W_{opt} is a matrix with orthonormal columns, consisting of the set of generalized eigenvectors of S_B and S_W corresponding to the m largest eigenvalues.

There are at most c-1 non-zero generalised eigenvalues, where c is the number of classes. The between-class scatter matrix S_B and the within-class scatter matrix S_W are defined as:

$$S_B = \sum_{i=1}^{c} N_i (\boldsymbol{m}_i - \boldsymbol{m})(\boldsymbol{m}_i - \boldsymbol{m})^T$$

$$S_W = \sum_{i=1}^{c} \sum_{\boldsymbol{x} \in C_i} (\boldsymbol{x} - \boldsymbol{m}_i)(\boldsymbol{x} - \boldsymbol{m}_i)^T$$

where $\boldsymbol{x}$ is a sample of class C_i, while $\boldsymbol{m}_i$ is the sample mean of class i, n_i is the number of samples in class i and $\boldsymbol{m}$ is the total sample mean.

QUESTIONS?

Exercises available on Moodle



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